6. Commercial Aircraft Operations



6. Commercial Aircraft Operations

6.1 Regulations and Scheduling

6.2 The Flight Deck

6.3 Flight Crew Activities During a Typical Flight

6. Commercial Aircraft Operations

6.1 Regulations and Scheduling

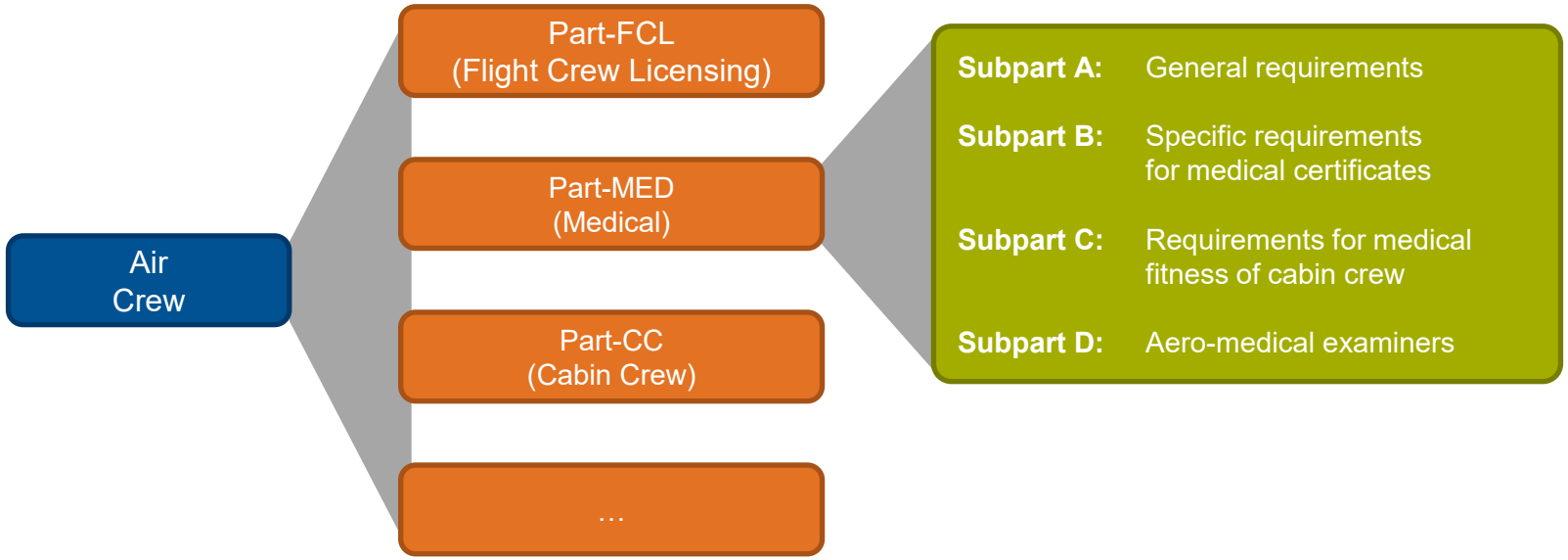


General Regulatory Authorities

Air carrier operations are generally regulated **by the country of registration** and the sovereignty **in which the operation is conducted**.

- USA: Federal Aviation Administration (FAA)
- EU: European Aviation Safety Agency (EASA)
- International flights: International Civil Aviation Organization (ICAO)

Example: EASA Air Crew Regulations



Flight Crew Regulation (1)

Flight Deck Crew

- Large carriers employ **5,000+ employees**.
- In many cases, crew members **may have never worked together** prior to a particular flight.
- Airline must **ensure safe, efficient, and smoothly functioning operations**.

➢ Very **detailed procedures** are developed, with only little room of improvisation

➢ **Operating Manuals + Checklists**

➢ The Captain has **Emergency Authority**



Flight Crew Regulation (2)

Cabin Crew

- Cabin Crew is primarily responsible for **passenger safety during flight**.
  - Other services: customer servicing, assistance with boarding, etc.

- Flight attendants receive a specialized training:
  - **Aircraft emergencies**
  - **Evacuation procedures**
  - **Medical assistance (incl. health hazards)**
  - **Care of special needs passengers**
  - **Flight regulations**
  - **Meal servicing**

- Cabin crews are required to undergo **annual recurrent training** for each of the aircraft on which they are qualified.



Crew Resource Management (CRM)

CRM addresses the effective use by the flight crews of **all resources available to them**:

- **People**: Flight attendants, dispatchers, maintenance personnel, ground/gate personnel, Air Traffic Control
- **Flight deck instrumentation**
- **Personal skills** and perception

CRM training emphasizes the skills necessary to optimize the human/machine interface including:

- **Situation awareness**
- **Use of automation systems**
- **Team building**
- **Task delegation**
- **Information transfer**
- **Problem solving & decision making**



Flight Crew Scheduling

Flight crews are usually assigned to a flight as part of their **normal schedules** (also referred to as "**rosters**").

A **roster**…

- describes a **crew member's flight activity for some period** (typically a month),
- consists of **sequences of flight duty and days off** (e.g., three days on, two days off)
- may range **from 65 to 100 hours spread over 10 to 18 duty days**

- A typical duty day consists of **one to four flight legs**, but on some days the crew may operate non at all, or fly more than five legs.

- Many departures require **the spontaneous filling of an "open seat"** because of a sick call, vacation, misconnect, legality issues, etc.
  - It is the responsibility of the airline's **crew scheduling department** to fill these open seats and satisfy the crew requirements for each flight.

6. Commercial Aircraft Operations

6.2 The Flight Deck



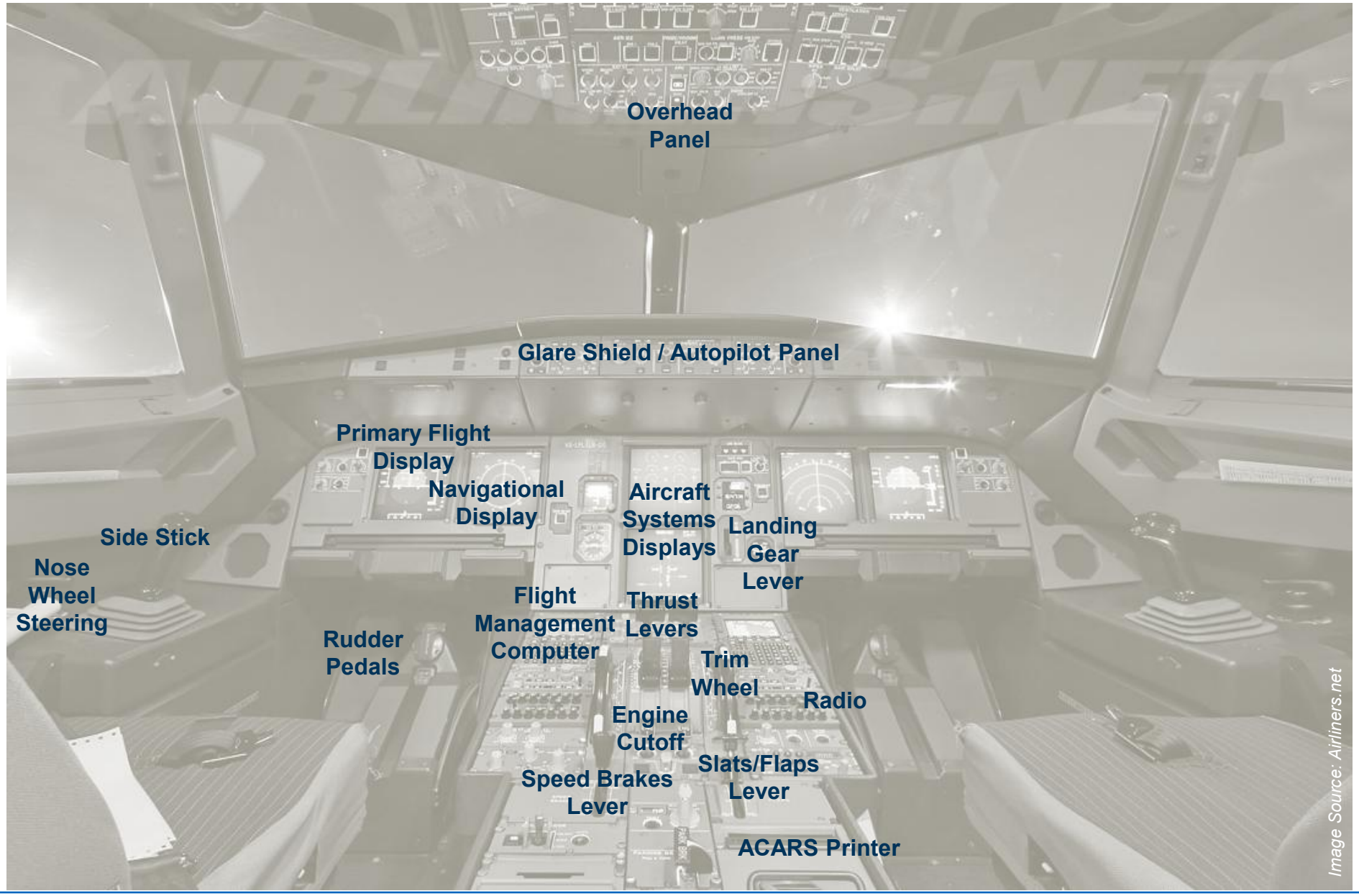
Overview: Airbus A320 (1)



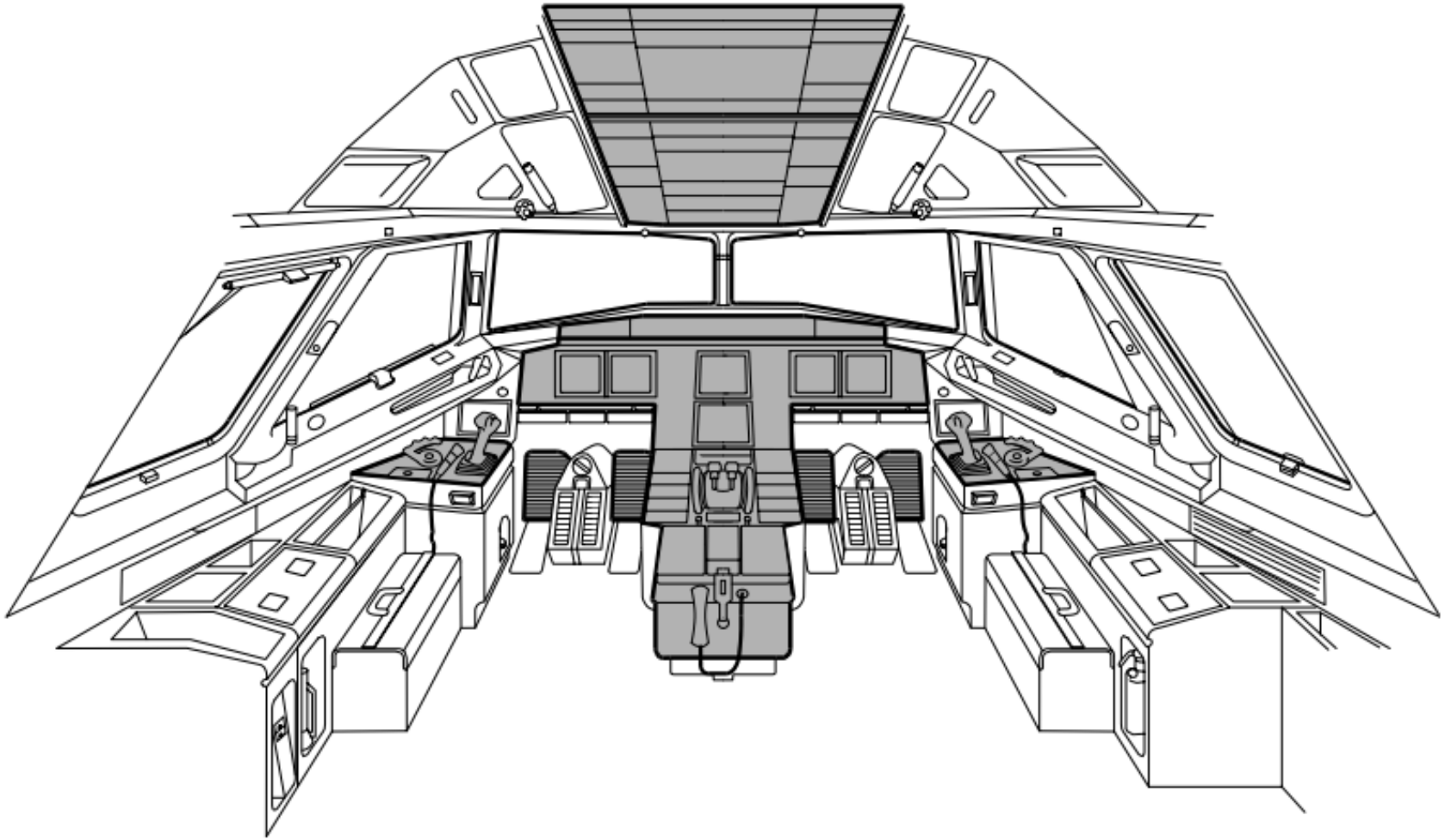
Overview: Airbus A350 (1)



Overview: Airbus A320 (2)

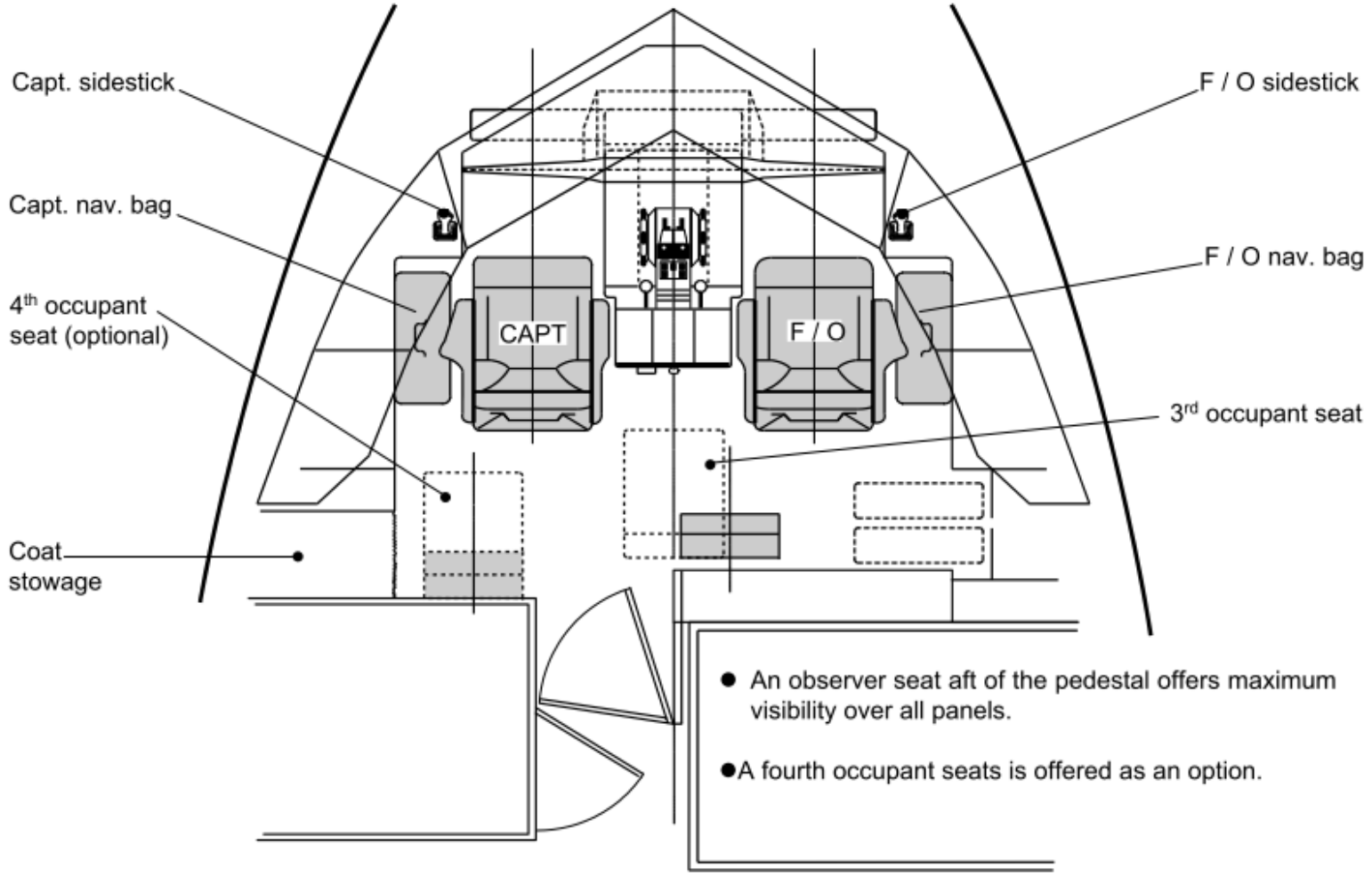


A319/A320/A321 Flight Deck – Main Features



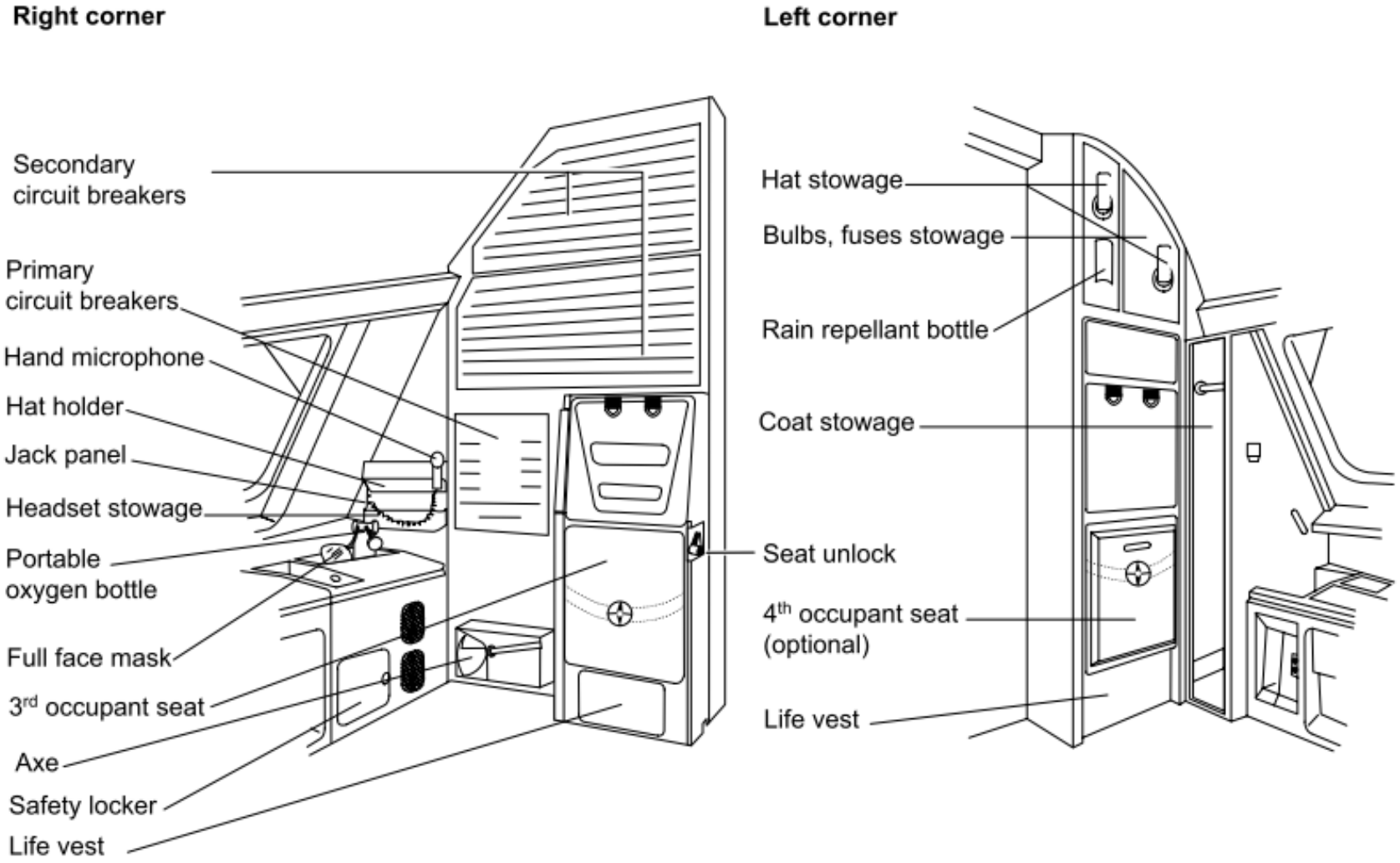
Source: Airbus

A319/A320/A321 Flight Deck – Plan View



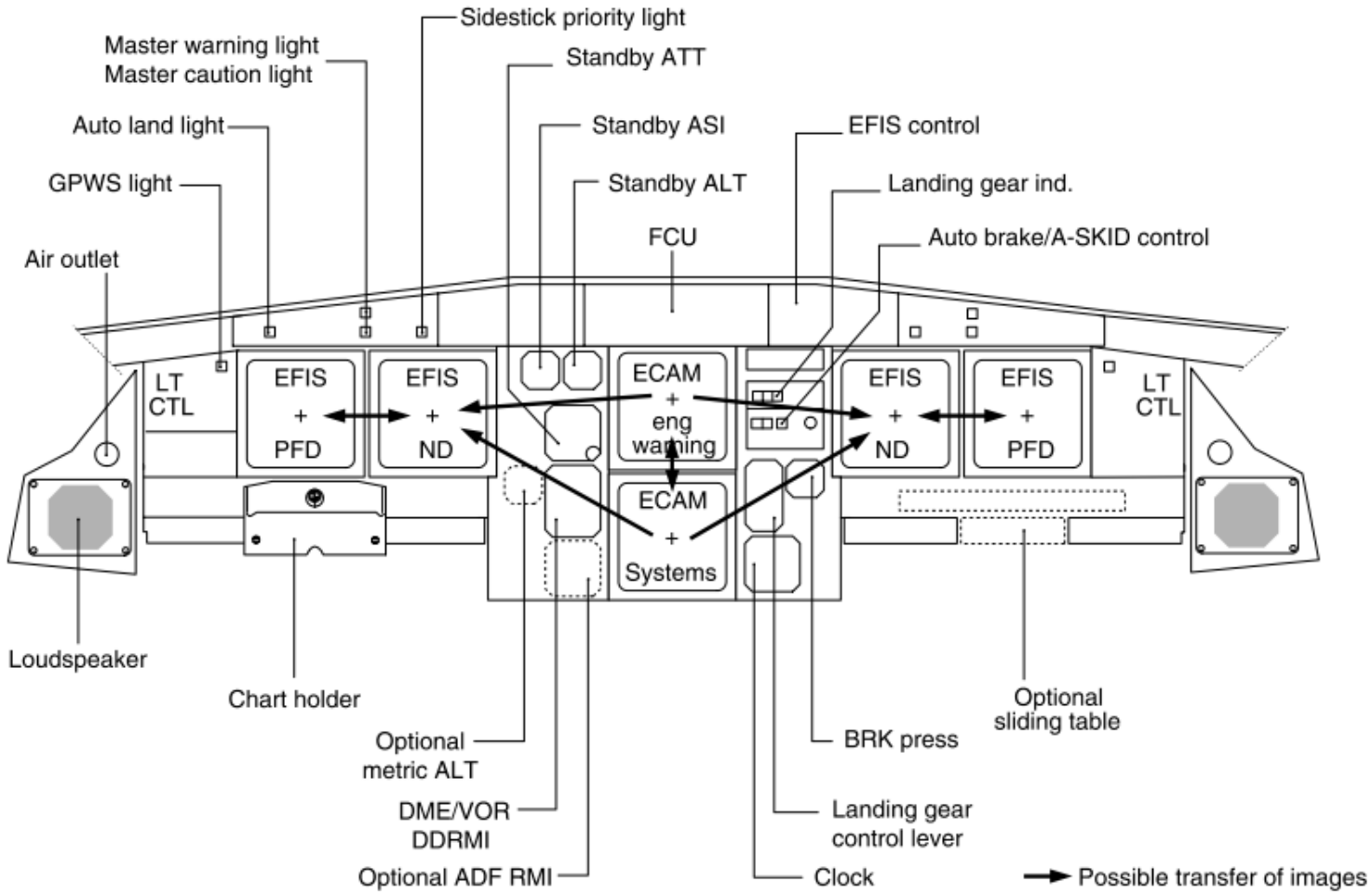
Source: Airbus

A319/A320/A321 Flight Deck – Rear View



Source: Airbus

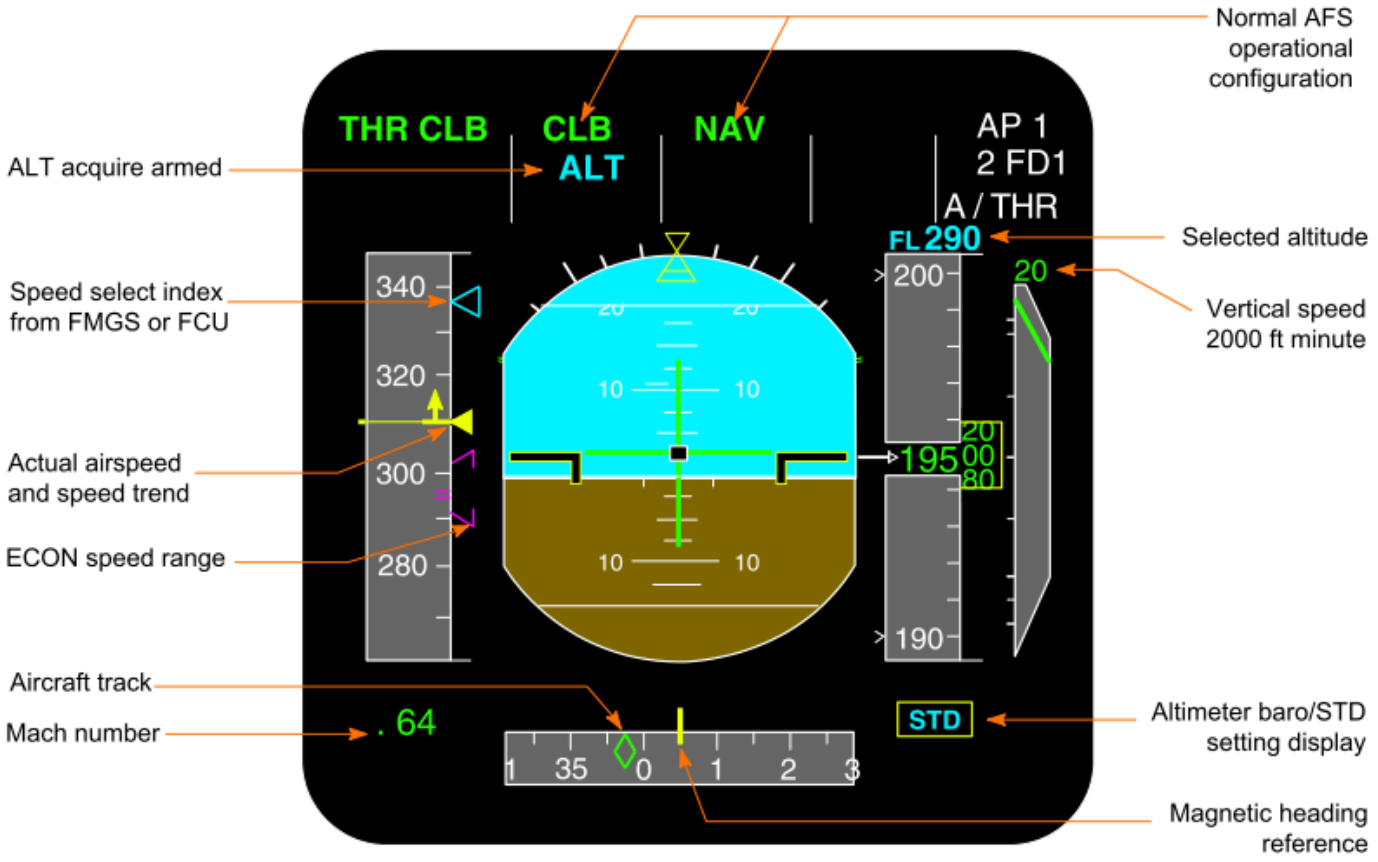
A319/A320/A321 Flight Deck – Main Instruments



Source: Airbus

A319/A320/A321 Flight Deck – Primary Flight Display (PFD) (1)

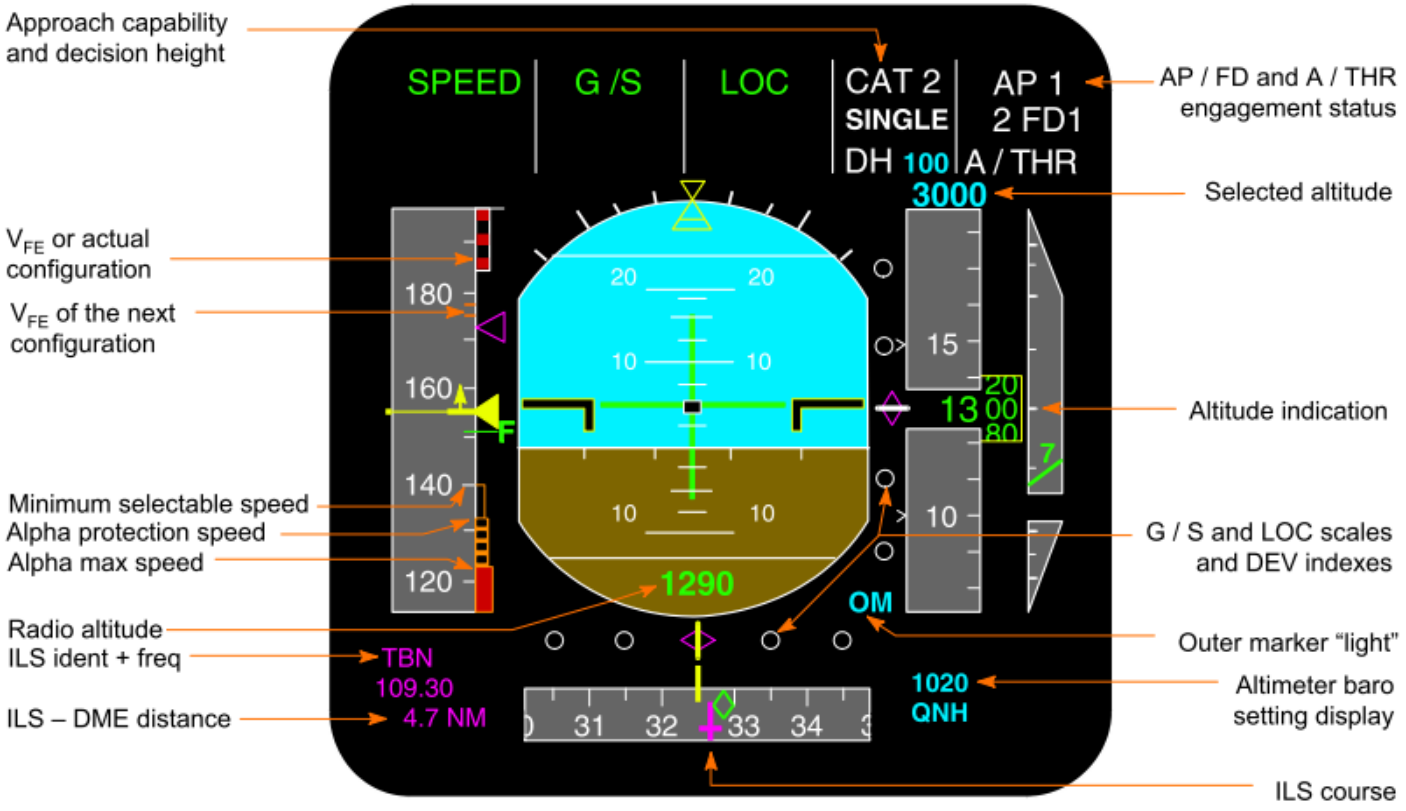
Climb Configuration



Source: Airbus

A319/A320/A321 Flight Deck – Primary Flight Display (PFD) (2)

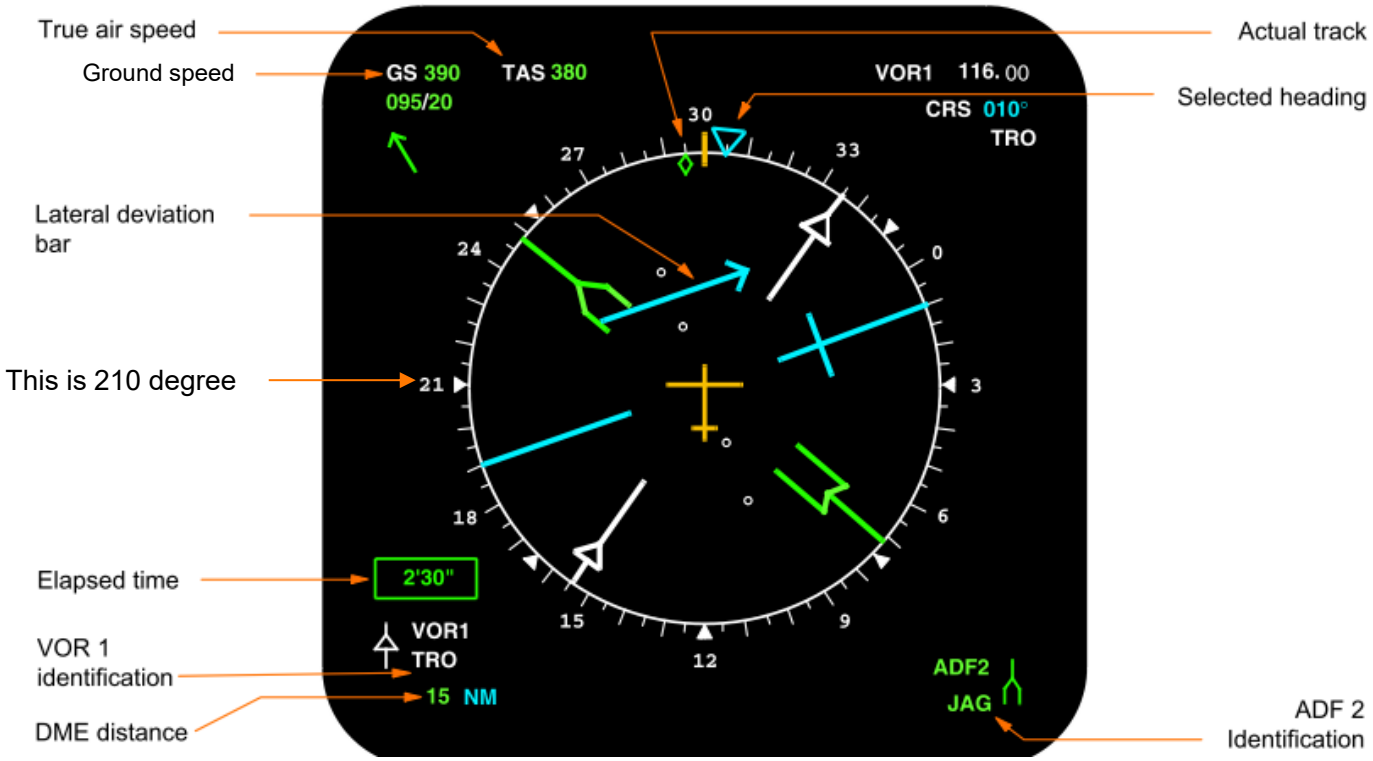
Approach Configuration



Source: Airbus

A319/A320/A321 Flight Deck – Navigational Display (ND) (1)

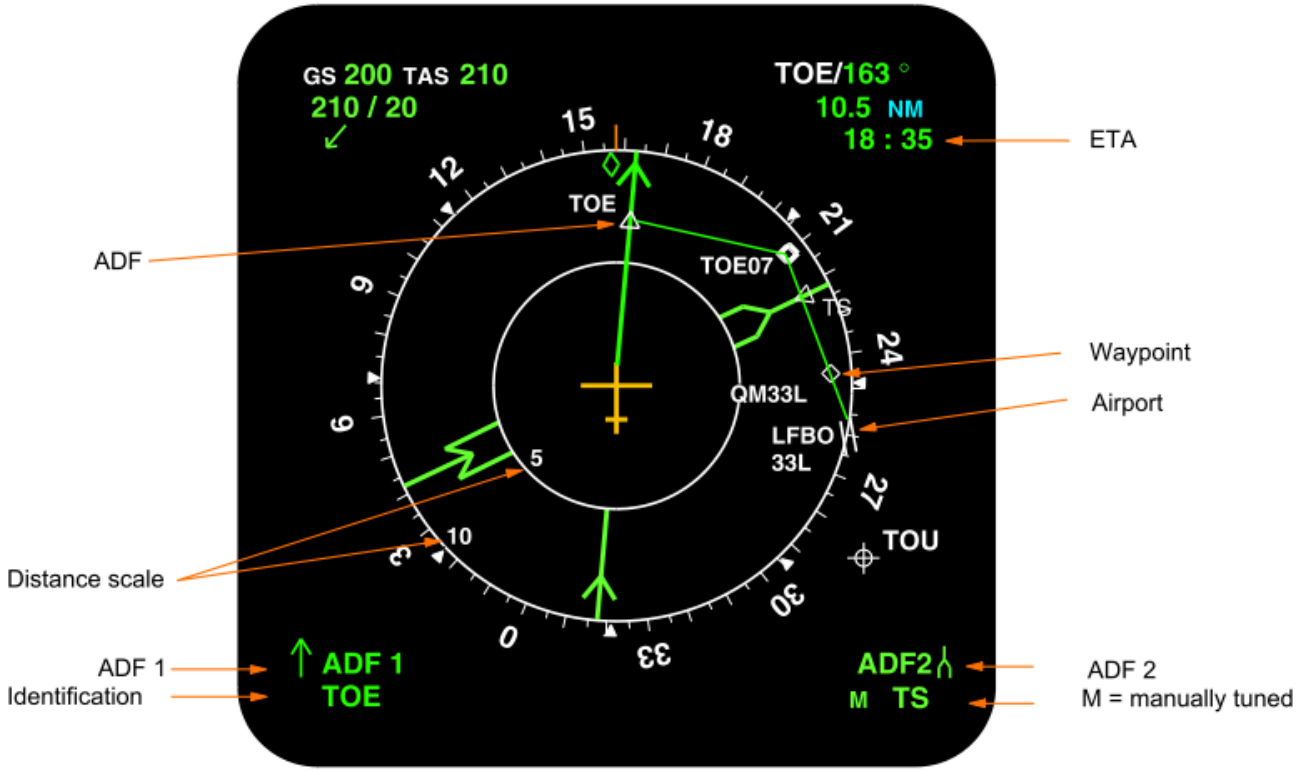
ROSE / VOR mode



Source: Airbus

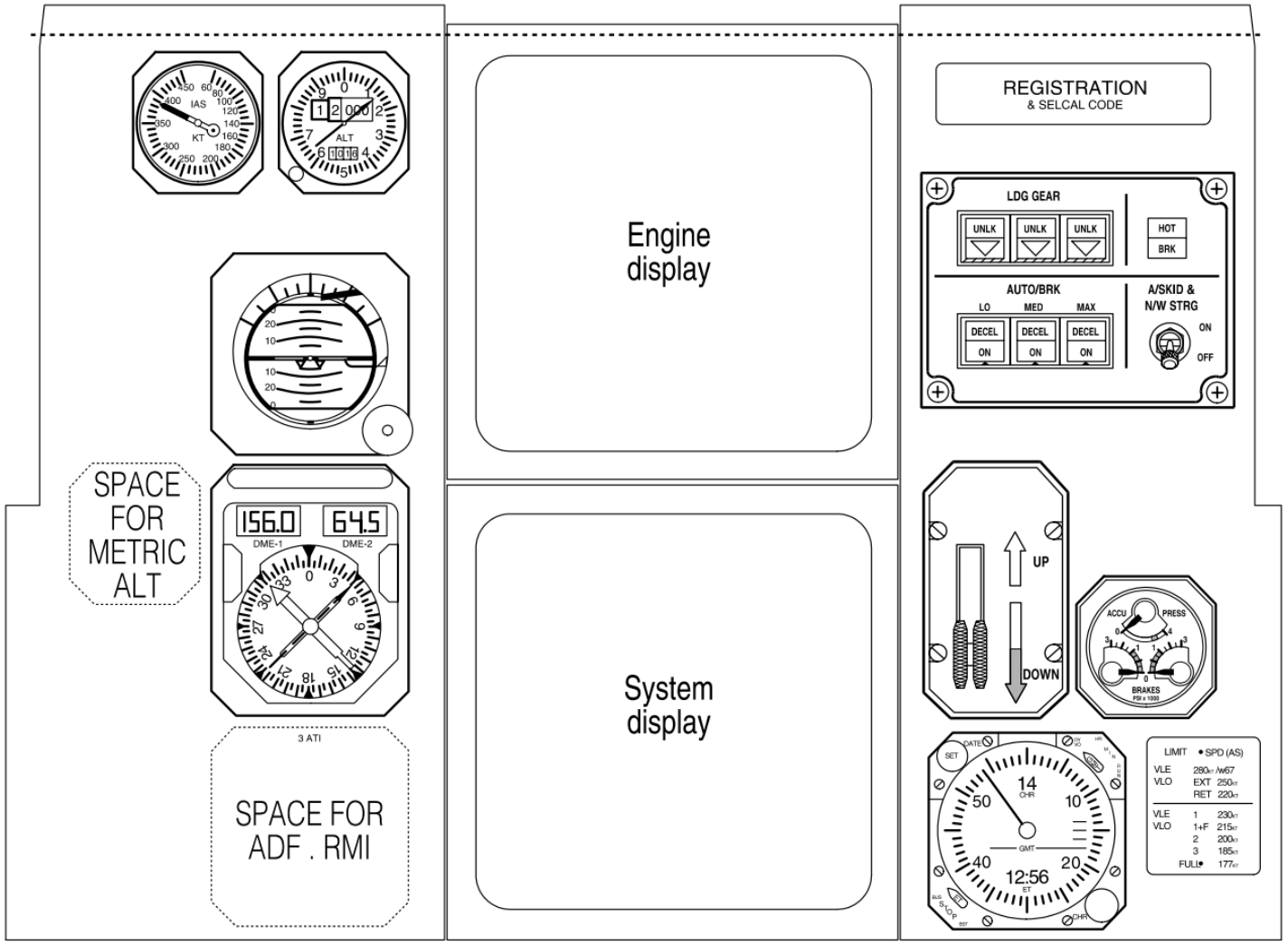
A319/A320/A321 Flight Deck – Navigational Display (ND) (2)

ROSE / NAV mode



Source: Airbus

A319/A320/A321 Flight Deck – Center Panel



Source: Airbus

A319/A320/A321 Flight Deck – ECAM* System Pages

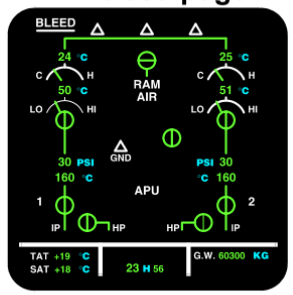
Cruise page



Engine page



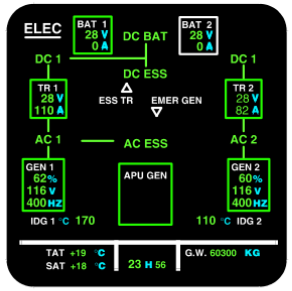
Air bleed page



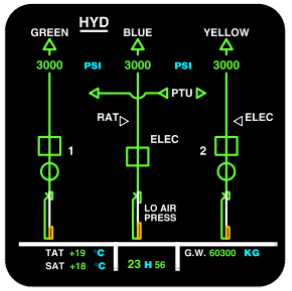
Cabin pressurization page



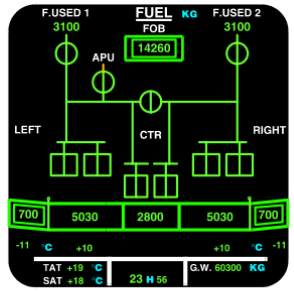
Electric page



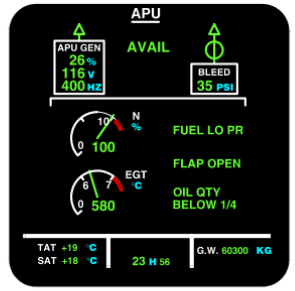
Hydraulic page



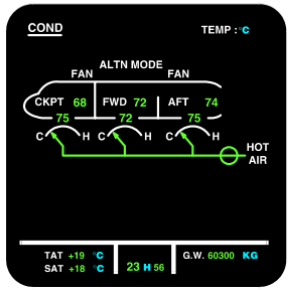
Fuel page



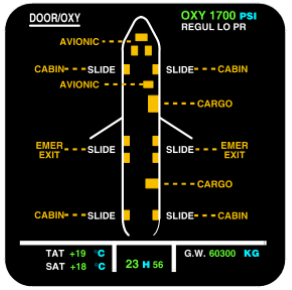
APU page



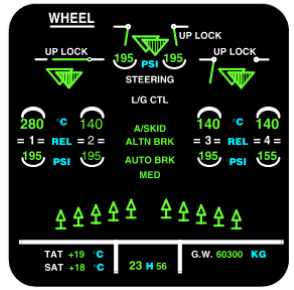
Air conditioning page



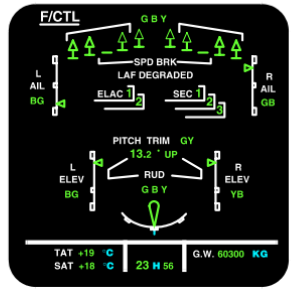
Doors/oxygen page



Wheel page



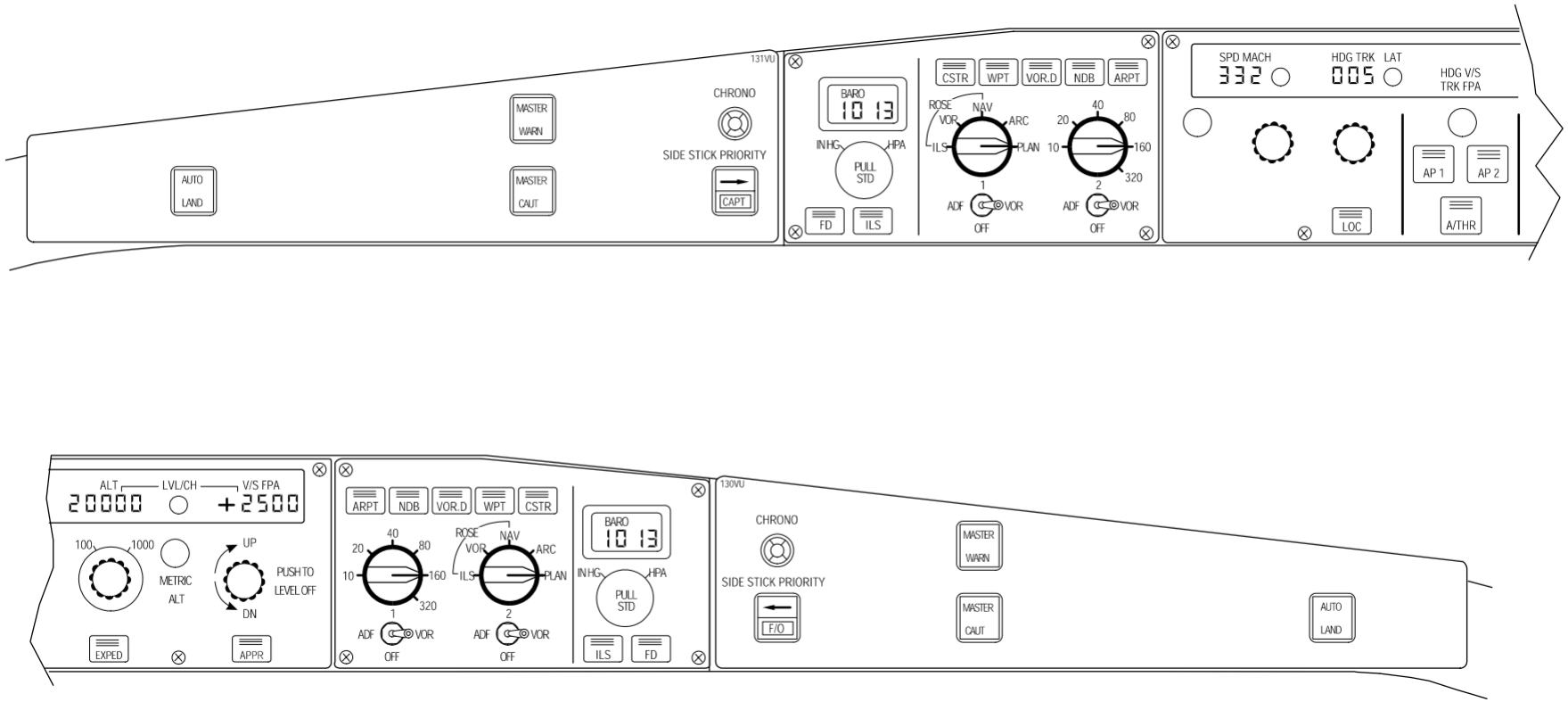
Flight controls page



** Electronic Centralized Aircraft Monitoring*

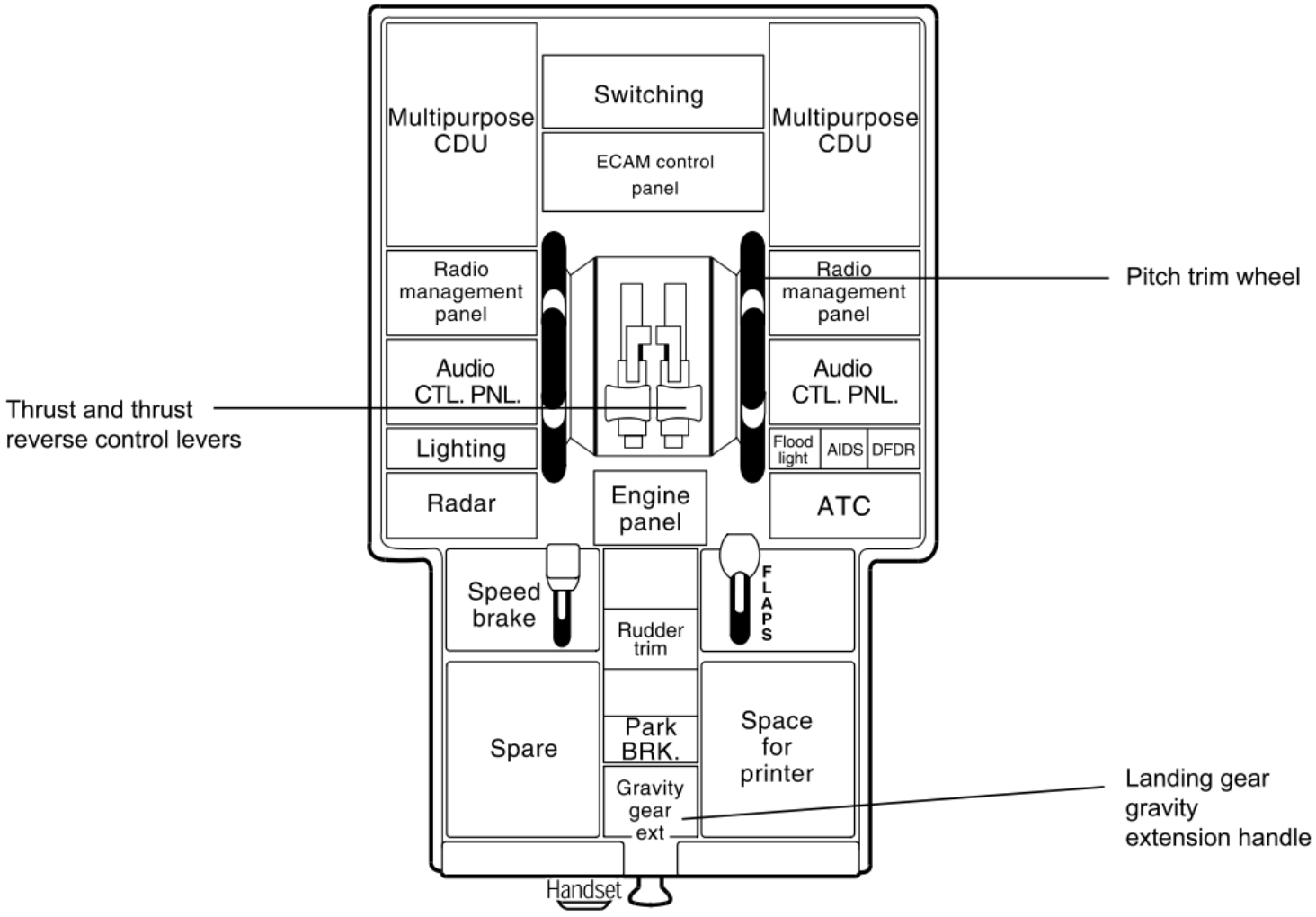
Source: Airbus

A319/A320/A321 Flight Deck – Glare Shield



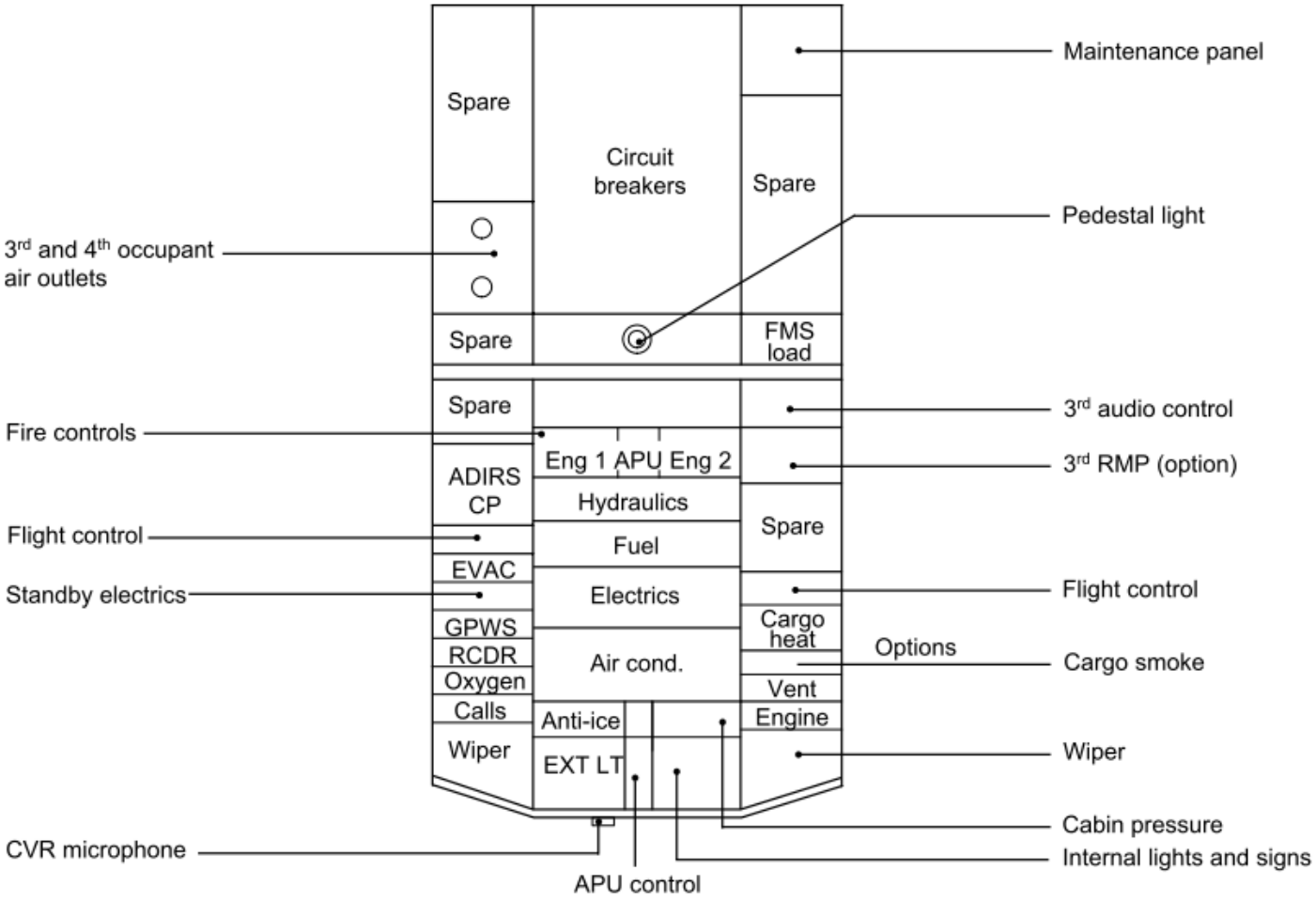
Source: Airbus

A319/A320/A321 Flight Deck – Pedestal



Source: Airbus

A319/A320/A321 Flight Deck – Overhead Panel



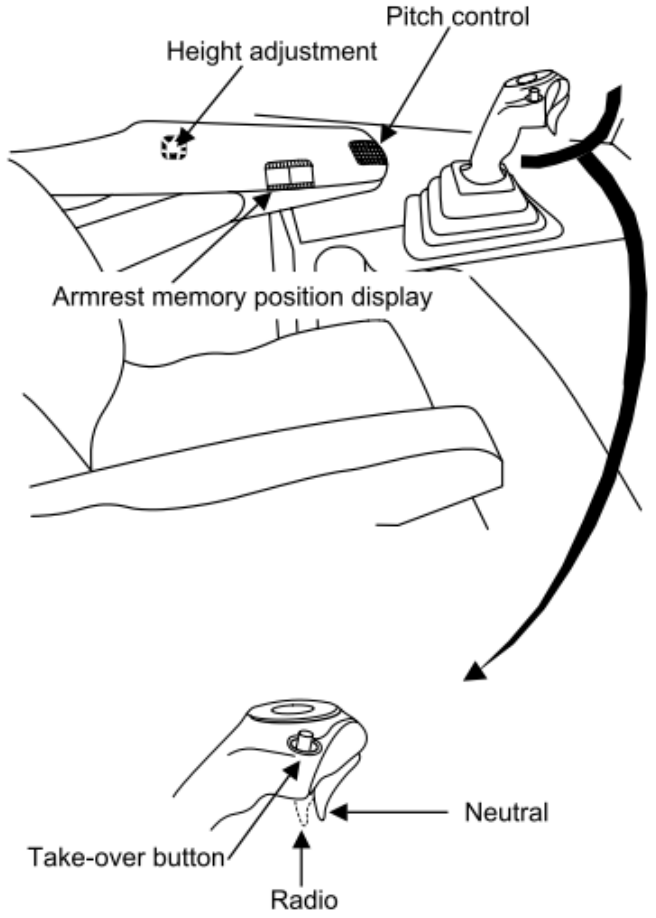
Source: Airbus

A319/A320/A321 Flight Deck – Side Stick (1)

- Sidesticks are installed on the CAPT and F / O forward lateral consoles.
- An **adjustable armrest** to facilitate free wrist movement is fitted on each seat.
- The sidestick works against a spring force proportional to the angular displacement.
- Sidestick includes :
  - radio communication trigger
  - a take-over button for autopilot disconnection and priority take-over.



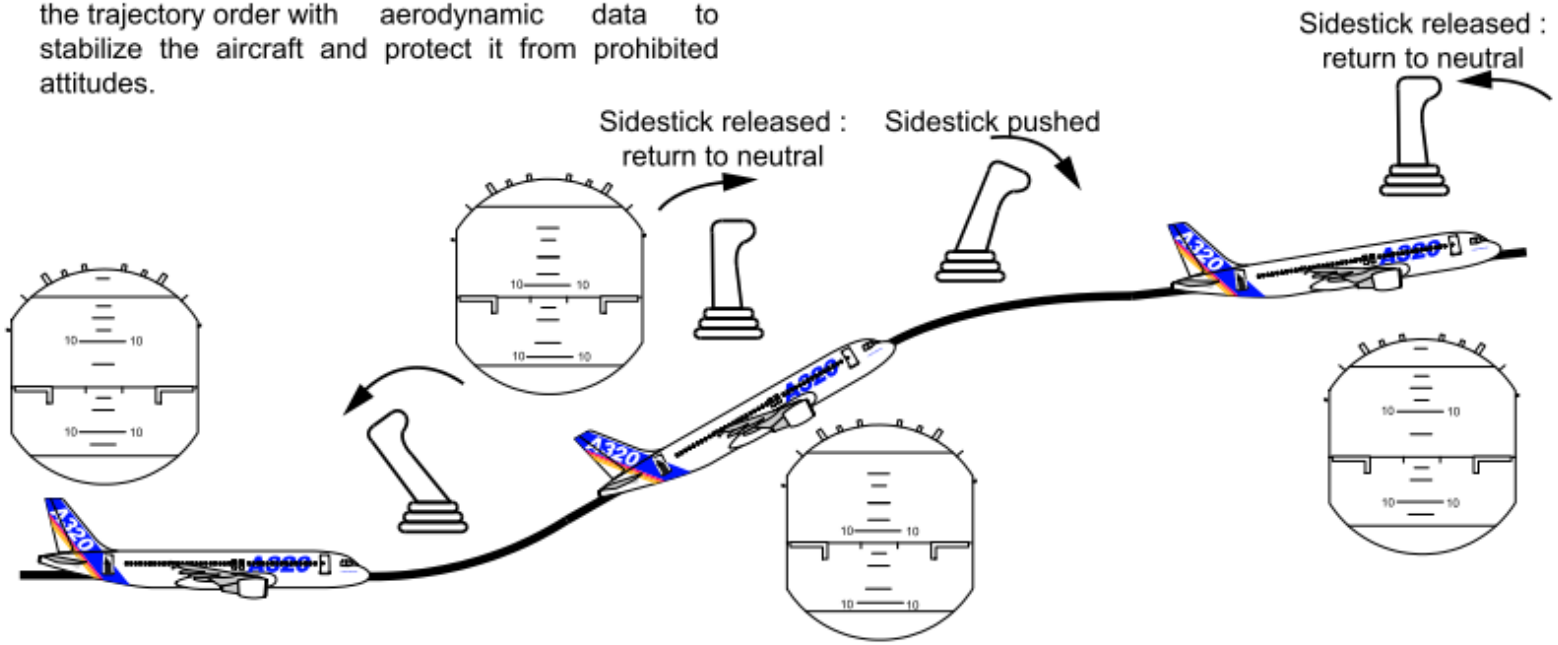
Image Source: Airliners.net



Source: Airbus

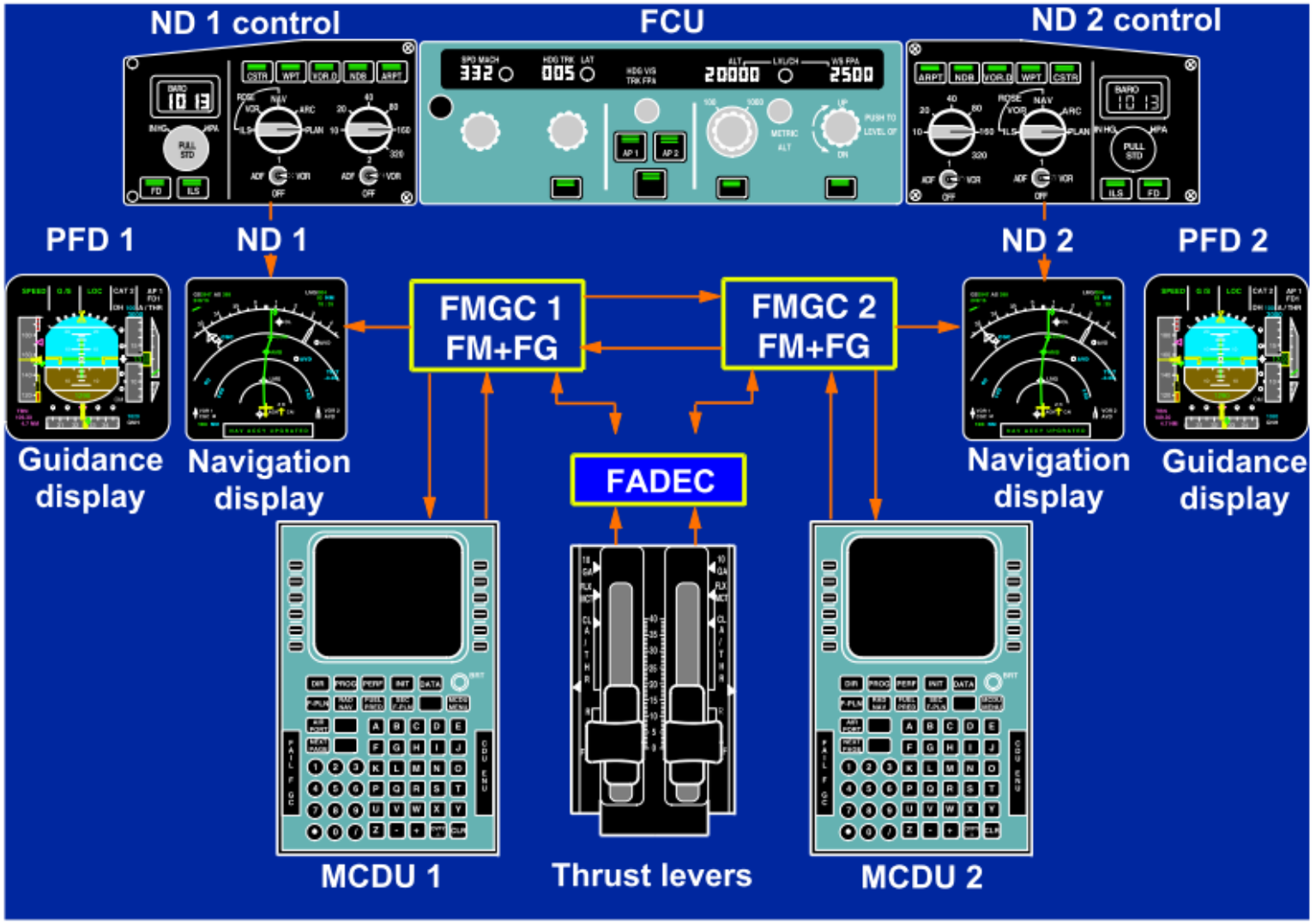
A319/A320/A321 Flight Deck – Side Stick (2)

- Moving the sidestick results in “setting the aircraft trajectory” with a certain level of “g” for the requested manoeuvre depending on the amount of sidestick movement.
- Movement is very precise since back lash and friction are negligible.
- Control of the flight path is performed by the Electronic Flight Control System **(EFCS)** which links the trajectory order with aerodynamic data to stabilize the aircraft and protect it from prohibited attitudes.



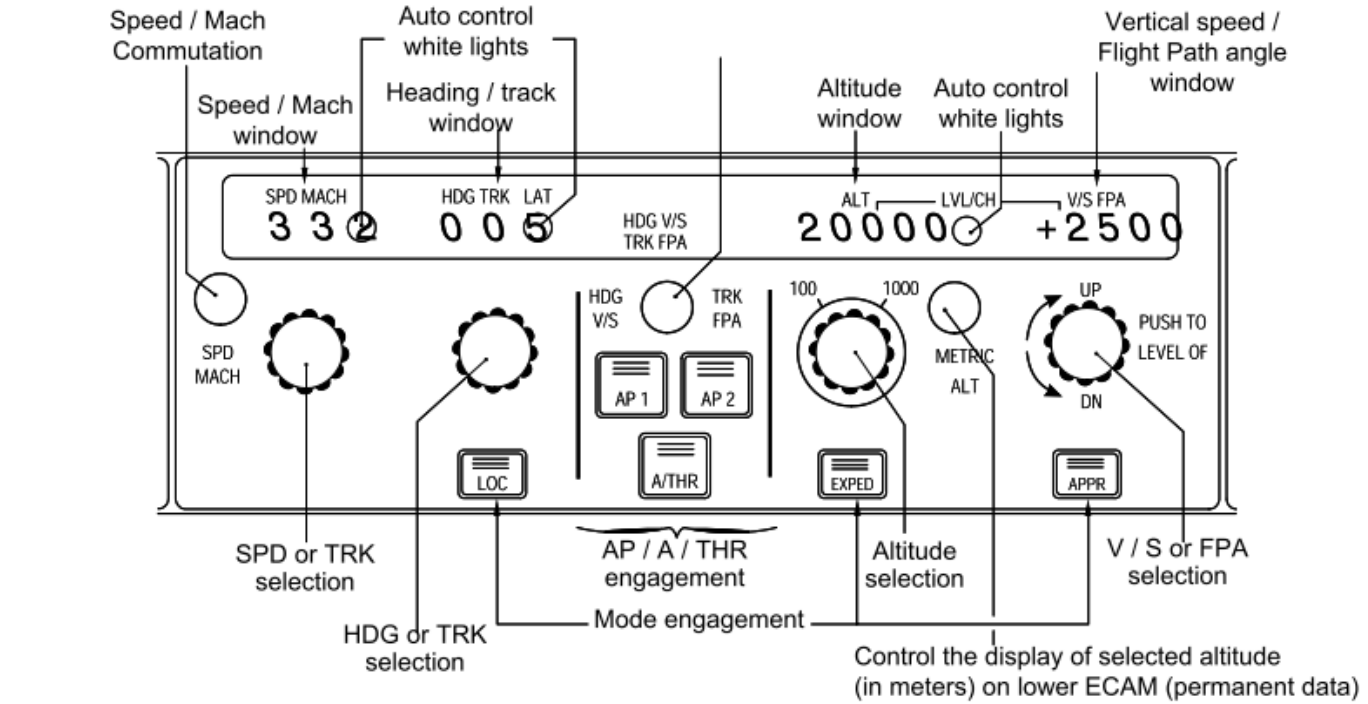
Source: Airbus

A319/A320/A321 Flight Deck – Flight Management and Guidance System (1)

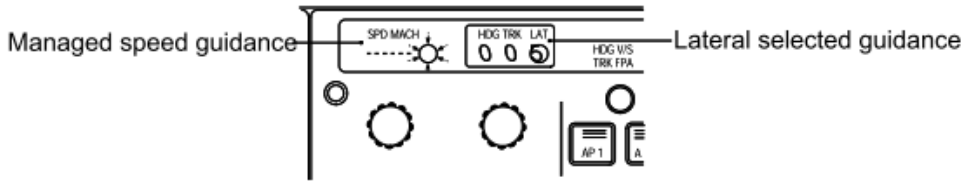


Source: Airbus

A319/A320/A321 Flight Deck – Flight Management and Guidance System (2)



- Managed guidance parameter window is dashed and auto control illuminated.



Source: Airbus

6. Commercial Aircraft Operations

6.3 Flight Crew Activities During a Typical Flight



Typical Flight Phases

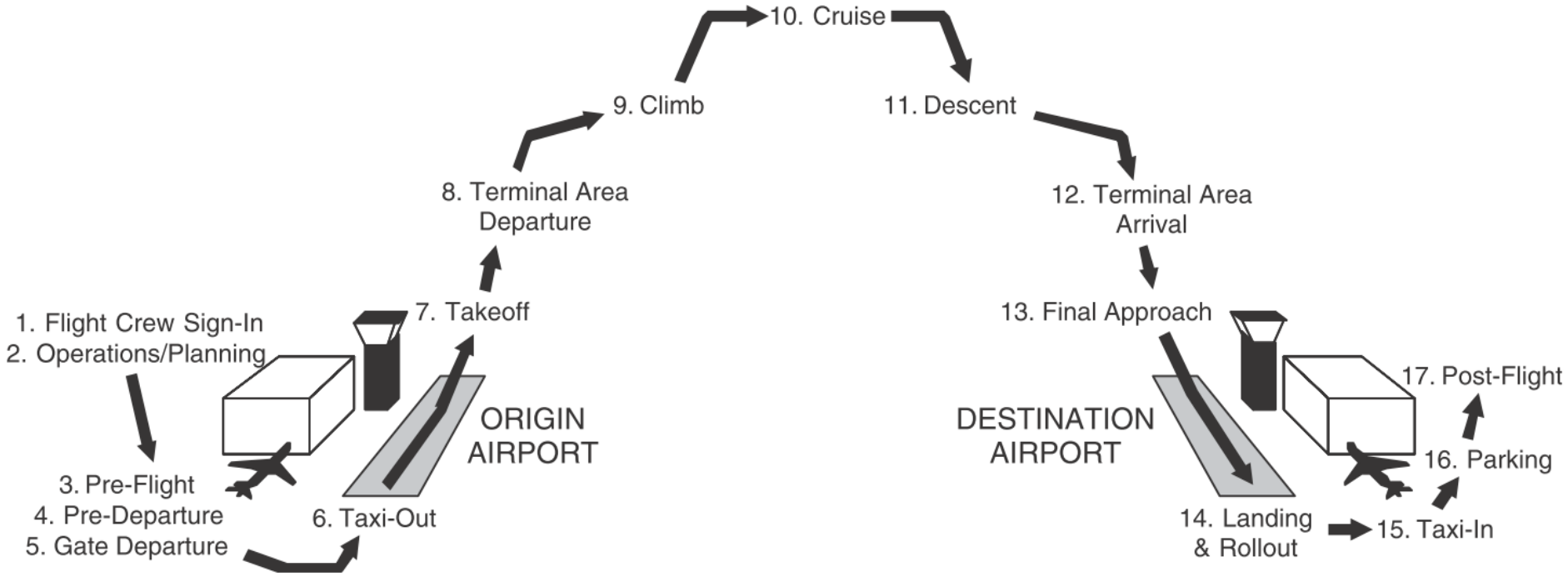


Image Source: Midkiff, Hansman, Reynolds (2009)

Example Flight

Route:

- Departure: Munich (MUC)
- Destination: Chicago (ORD)





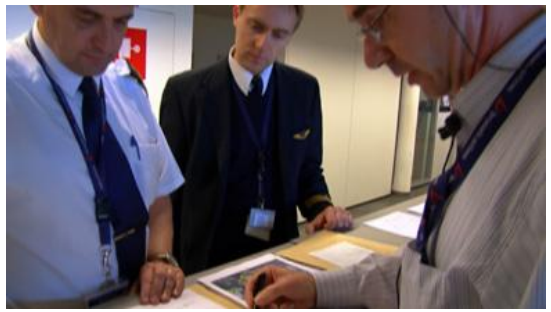
Aircraft Type:

- Airbus A350-900
- Engines: Trent XWB
- MTOM: 283,000 kg
- Payload: 53,300 kg



1. Flight Crew Sign-In

- Crew members are usually required to **sign in at the airport flight operations office** 1 hour prior to the departure.
- The flight crew starts the **flight planning tasks**.
- The **first officer** may proceed to the aircraft earlier than the captain to **begin pre-flight duties**, while the **captain completes the necessary paperwork** in the flight operations office.

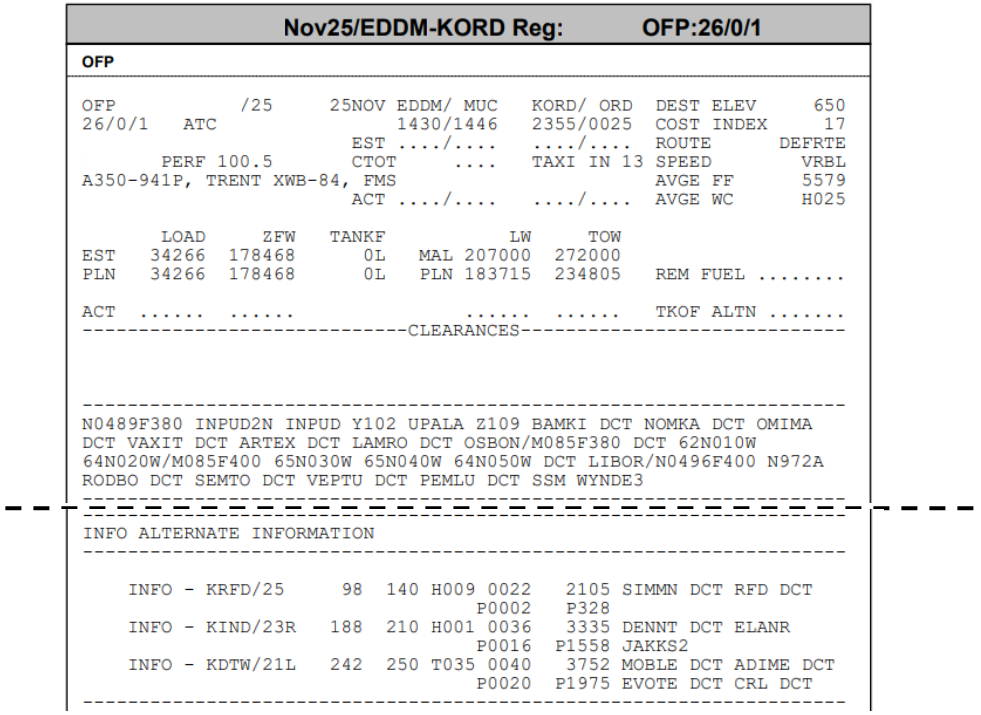


2. Operations / Planning

- Most airlines have a central Airline Operations Control Center (AOCC) staffed by certified **flight dispatchers**.
- The **flight planning tasks** include
  - ✓ selecting the best routing (mission time, fuel burn, external conditions)
  - ✓ and generating a flight plan
- A **flight plan** accounts for
  - ✓ aircraft type
  - ✓ forecast weather conditions
  - ✓ aircraft performance and loading
  - ✓ aircraft mechanical conditions
  - ✓ marketing constraints
  - ✓ airport restrictions
  - ✓ company policies (e.g., fuel vs. time)

Source: Midkiff, Hansman, Reynolds (2009)

Flight Plan (1)



Nov25/EDDM-KORD Reg: OFP:26/0/1

OFP

```
OFP           /25     25NOV EDDM/ MUC   KORD/ ORD  DEST ELEV     650
26/0/1   ATC                1430/1446   2355/0025  COST INDEX     17
                          EST ..../....   ..../....  ROUTE      DEFRTE
       PERF 100.5         CTOT     ....   TAXI IN 13 SPEED        VRBL
A350-941P, TRENT XWB-84, FMS                        AVGE FF       5579
                          ACT ..../....   ..../....  AVGE WC       H025

       LOAD     ZFW   TANKF           LW     TOW
EST   34266  178468      0L   MAL 207000  272000
PLN   34266  178468      0L   PLN 183715  234805   REM FUEL ........

ACT  ......  ......                ......  ......   TKOF ALTN .......
------------------------------CLEARANCES-----------------------------



--------------------------------------------------------------------------
N0489F380 INPUD2N INPUD Y102 UPALA Z109 BAMKI DCT NOMKA DCT OMIMA
DCT VAXIT DCT ARTEX DCT LAMRO DCT OSBON/M085F380 DCT 62N010W
64N020W/M085F400 65N030W 65N040W 64N050W DCT LIBOR/N0496F400 N972A
RODBO DCT SEMTO DCT VEPTU DCT PEMLU DCT SSM WYNDE3
--------------------------------------------------------------------------
INFO ALTERNATE INFORMATION
--------------------------------------------------------------------------

    INFO - KRFD/25     98  140 H009 0022   2105 SIMMN DCT RFD DCT
                                 P0002    P328
    INFO - KIND/23R   188  210 H001 0036   3335 DENNT DCT ELANR
                                 P0016   P1558 JAKKS2
    INFO - KDTW/21L   242  250 T035 0040   3752 MOBLE DCT ADIME DCT
                                 P0020   P1975 EVOTE DCT CRL DCT
--------------------------------------------------------------------------
```

Header

General information: aircraft type, engine type routing information, time constraints, cost index

Flight plan route: departure airport, SID, waypoints, airways, STAR, destination airport…

Detailed routing: …distance information, cruising altitude, airspeeds, planned fuel

Alternate airport information: alternate airport names, distance information, cruising altitudes, trip time, required fuel, routing information

Flight Plan (2)

```
TRIP       51090  09.09

CONT DES    1169  00.13  CYYZ WXFM 2235 WXTO 0035

ALTN        1777  00.20  KMKE WXFM 2316 WXTO 0116

FINAL RES   2301  00.30

PLNTOF     56337  10.12                          FIVE SIX THREE

DIFF      ......  .....

PLNTOF    ......  .....  (CORRECTED)

TAXI       ....           400/00.16

MINBLOCK  ......         TCAP 112636  .....  REM FUEL ...... AT ....

FUEL OB   ......  .....

DISCF     ......  .....  POSS  23285L

BLOCK     ......         TCAP 112636  .....  REM FUEL ...... AT ....
--------- --------- --------- --------- --------- --------- --------
CONT 3%     1533 00:16 AT DEP
(364 KG INCLUDED IN TRIP)
--------- --------- --------- --------- --------- --------- --------
ZFW CORR PS 1000 PLNTOF PS 239  / MS 1000 PLNTOF MS 228
2000 FT BELOW                   TRIP PS 989  / TIME PS 0.01
4000 FT BELOW                   TRIP PS 2287 / TIME PS 0.12
2000 FT ABOVE                         NOT AVAILABLE
ECON CI100                      TRIP PS 177  / TIME MS 0.03
```

Fuel Information: trip fuel, minimum contingency fuel, alternate fuel, final reserve fuel, planned block fuel, extra fuel

Statistical information about mean value of fuel loading for this flight mission

Data summary

Fuel

One of the key factors in flight planning is determining the fuel load. The following must be taken into account:

- TRIP: fuel needed to reach the **destination airport**
- **CONT: reserve fuel** (may vary depending on type of flight, weather conditions, etc.)
- ALTN: destination **weather** and alternates
- FINAL RES: for 30 additional minutes in holding pattern
- TAXI
- EXTRA (pilot discretion)
- cruising **speed and altitude** (driven by the company's policies, constrained by ATC)
- ferrying fuel (**tankering**)
- required **takeoff and landing performance**
- **payload** to carry



Source: Midkiff, Hansman, Reynolds (2009)

Takeoff Planning

Takeoff performance must be evaluated once **the load is determined** and **available runways considered**.

- **Takeoff planning considerations** also include: maximum runway* and climb limited weights**, takeoff engine out performance, takeoff abort procedures, and obstacle clearance
- Planned takeoff data is usually calculated by the **flight crew**.
- **Adjustment to takeoff plans** may have to be made to deal with weather changes, weight penalties, or runway contamination (e.g., snow, slush, water)

* The **runway limited weight** is derived from the most restrictive performance based on runway length, slope, obstacle clearance, brake energy, and tire speed.

** The **climb limited weight** is independent of the runway used, and is based on the ability of the aircraft to climb at minimum angles with and without all engines operating normally.



Source: Midkiff, Hansman, Reynolds (2009)

Takeoff Performance: V-Speeds

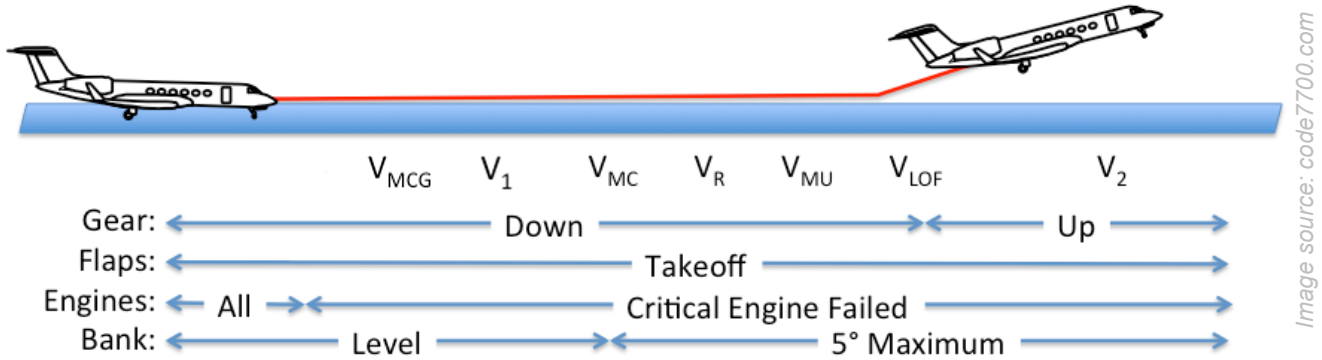


Image source: code7700.com

- **V_{MCG} Minimum control speed on the ground.** At this speed, the aircraft must be able to continue a straight path down the runway with a failed engine, without relying on nose gear reactions.
- **V_1 Decision speed**, a short time after critical engine failure speed. Above this speed, aerodynamic controls alone must be adequate to proceed safely with takeoff.
- **V_{MC} Minimum Control Speed.** Minimum airspeed at which when critical engine is made inoperative, it is still possible to recover control of the airplane and maintain straight flight.
- **V_R Rotation Speed.** Must be greater than V_1 and greater than 1.05 V_{MC}
- **V_{MU} Minimum Unstick Speed.** Minimum airspeed at which airplane can safely lift off ground and continue take-off.
- **V_{LOF} Lift-off Speed.** Must be greater than 1.1 V_{MU} with all engines, or 1.05 V_{MUC} with engine out.
- **V_2 Take-off climb speed** is the demonstrated airspeed at the 35 ft height. Must be greater than 1.1 V_{MC} and 1.2 V_s (stalling speed in takeoff configuration).

Source: Aircraft Aerodynamics and Design Group Stanford

Cabin Crew Briefing

Communication between the cockpit and cabin crew members is critical to the safety and efficiency of the flight.

The **cabin crew briefing** (led by the captain) includes:

- En-route flight time and destination weather
- Taxi-out time
- Security issues and alerts
- Ride conditions and expected turbulence
- Inoperative cabin components
- Crew meal services
- Adherence to sterile cockpit periods
- Any other safety-relevant and operational issues



Source: Midkiff, Hansman, Reynolds (2009)

3. Pre-Flight

The crew must determine **the airworthiness of the aircraft** and address any open issues before departure.

Pre-flight inspections include:

- **exterior walk-around examination:** visual inspection in which the crew checks for obvious damage to the fuselage, engines, wings, flight control surfaces, tire wear and pressure, brake wear indicators, absence of leaks, conditions of antennas, probes and lights, necessity for deicing, other factors that may affect flight safety
- **interior cockpit setup:** e.g., programming the Flight Management Computer
- **systems checks:** e.g., testing the fire warnings

Pre-flight inspections are formally established by **checklists**.



Source: Midkiff, Hansman, Reynolds (2009)

Autoflight Programming

Autoflight initialization and FMS (flight management system) programming are conducted during the pre-flight phase.

There are three automation levels regarding autoflight:

1. **Manual control** (hand flying)
2. **Tactical modes** (directly dialing in-flight parameters such as heading, altitude, airspeed)
3. **Strategic lateral and vertical navigation modes** (the aircraft is flown automatically by the autopilot along a flight path programmed into the FMS)

- Although the required trajectory parameters are usually programmed into the automation, **use of the autopilot is often discretionary**.
- **Takeoff** is always performed manually by the pilot.
- Depending on the approach procedure and weather conditions, **landings** are also frequently performed manually by the pilot.



Source: Midkiff, Hansman, Reynolds (2009)

Fuel Slip and Other Documents

To **confirm the fuel quantity and distribution** (between different fuel tanks) with the flight plan and on-board sensors, a fuel slip is provided to the crew by the fueler (airport ground staff).

Other required flight documents may include:

- Water and lavatory servicing verification
- Security inspection confirmation
- Documents accompanying restricted cargo (e.g., hazardous materials)
- Armed passenger (e.g., law enforcement personnel)
- Customs declaration (for international flights)
- Jumpseat rider identification confirmation



Source: Midkiff, Hansman, Reynolds (2009)

4. Pre-Departure

At approx. 20 minutes before departure, the pilots…

- …finalize the FMS and autoflight programming.
- …update their weather and airport information (ATIS)
- …obtain route clearance from ATC (which may differ from the filed routing: changes must then be addressed and reprogrammed (fuel, performance,…)

At approx. 5-10 minutes before pushback, the pilots…

- …turn on the “Fasten Seat Belt” signs which signal the cabin crew to ready the cabin for departure and deliver the requisite public address (PA) announcements
- …the captain conducts the **takeoff briefing**:
  - ➢ pilot flying / pilot non-flying decision, initial heading, altitude and departure procedure requirements (SID), obstacle clearance, noise abatement procedures, airport elevation, clean-up altitude, takeoff abort procedures (one engine out)

- Once all passengers are on board and seated, the ramp agent coordinates **closing the aircraft doors** with the captain and purser.
- The airport ground staff completes baggage and cargo loading, and **closes the cargo doors**.
- A tug is connected to the aircraft via a towbar (for **push-back procedures**).

Source: Midkiff, Hansman, Reynolds (2009)

4. Pre-Departure (ctd.)



Prior pushback:

- …the **load closeout** is received by ACARS that typically includes finalized aircraft and fuel weights, stabilizer trim settings (Airbus: after ENG START), center of gravity data, passenger count, cargo loading, live animal and security information.
  - The first officer uses the updated information to calculate finalized takeoff performance data.
  - The first officer also resets the stabilizer trim and set takeoff reference speeds.
  - Any weather changes also have to be taken into account.
- …performance calculation finalized
- …perform the “Prior To Push-Back” checklist
- …perform the “Before Engine Start” checklist

Source: Midkiff, Hansman, Reynolds (2009)

5. Gate Departure



Usually, the following procedure is happening to leave the gate:

- Move the **jetbridge** out of the way.
- (**De-ice** the aircraft either directly at the gate or at a designated area.)
- Remove wheel chocks from aircraft and release parking brake (**begin of Block Time**).
- Call ATC to **receive push-back clearance**.
- Perform push-back.
- **Start the engines** when the push-back crew signals the area is clear.

Once the aircraft has **reached its final position** after push-back,

- the towbar is disconnected and the parking brakes set again,
- a final confirmation is provided to the flight crew by the push-back crew that the ramp area is clear to taxi,
- ATC taxi clearance is received,
- the pilots check the outside area and announce “clear left” and “clear right,” respectively.

Source: Midkiff, Hansman, Reynolds (2009)

Example Flight

Gate Departure / Pushback at MUC from Stand 216, heading southwards to RWY 26L

MUC Apron 2

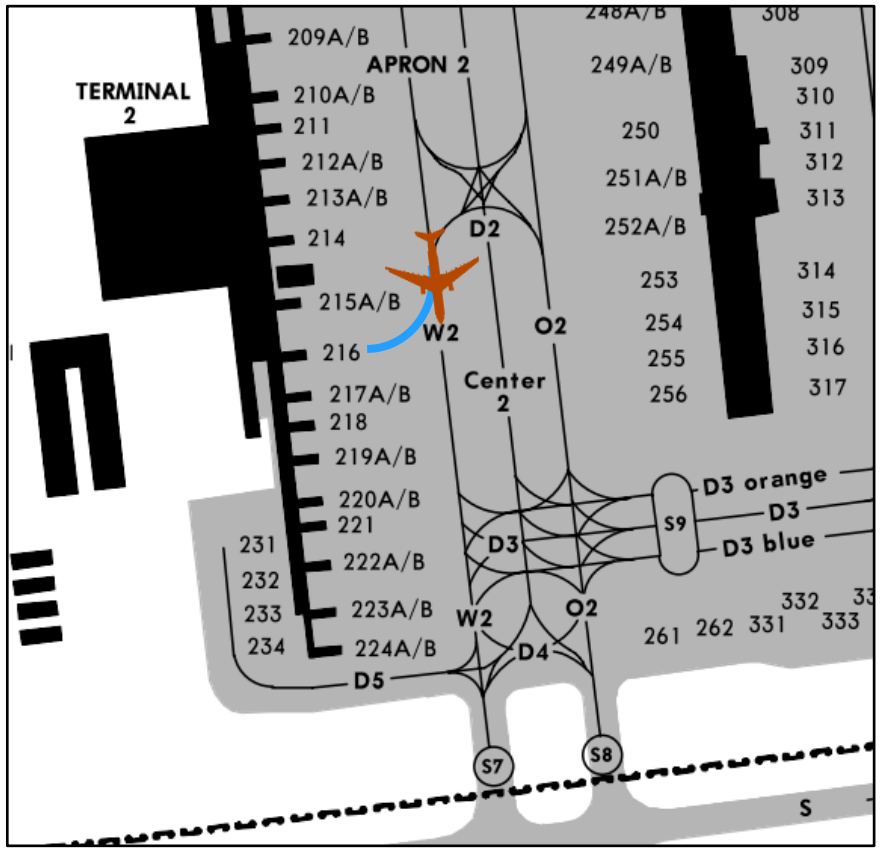




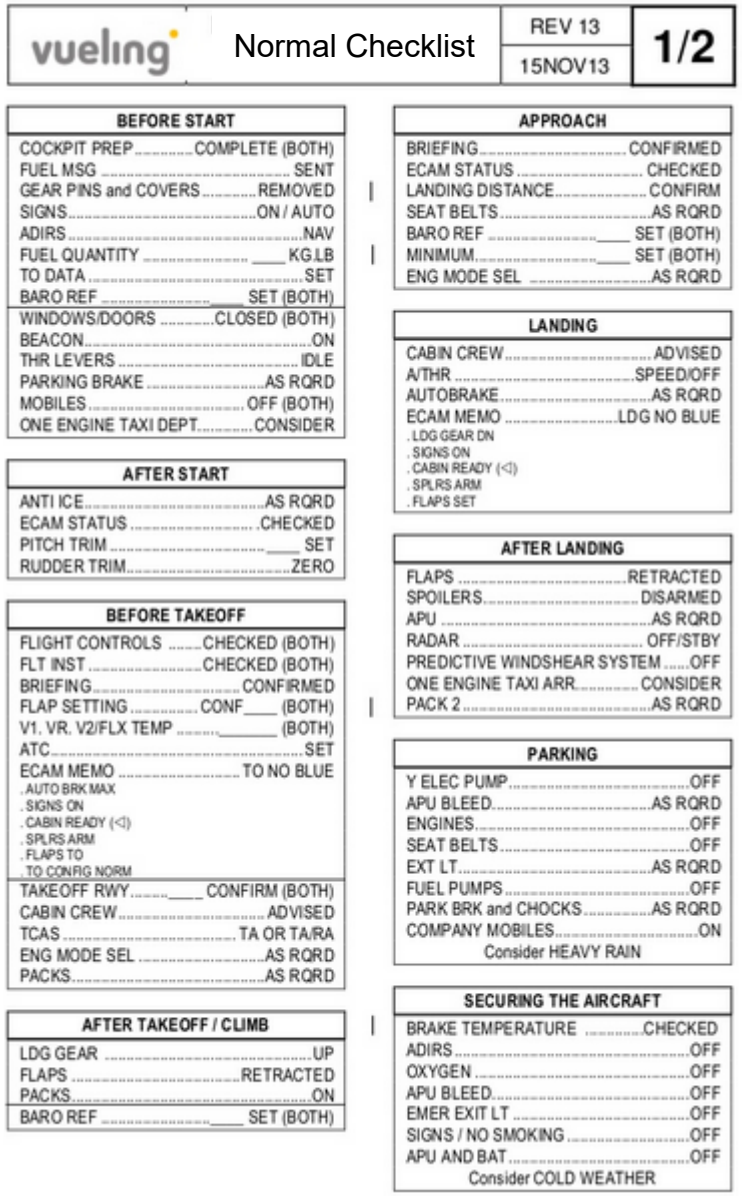
Image Source: Jeppesen

6. Taxi-Out

Once the aircraft has received taxi clearance, and the pilots have visually confirmed clearance of the area, the **aircraft can taxi to the departure runway**.

During taxi…

- The cockpit crew performs the **“Taxi” and “Before Takeoff” checklists**
- The captain must finally ensure **the cabin is ready** for takeoff.



vueling — Normal Checklist — REV 13 — 15NOV13 — 1/2

BEFORE START

COCKPIT PREP	COMPLETE (BOTH)
FUEL MSG	SENT
GEAR PINS and COVERS	REMOVED
SIGNS	ON / AUTO
ADIRS	NAV
FUEL QUANTITY	____ KG LB
TO DATA	SET
BARO REF	____ SET (BOTH)
WINDOWS/DOORS	CLOSED (BOTH)
BEACON	ON
THR LEVERS	IDLE
PARKING BRAKE	AS RQRD
MOBILES	OFF (BOTH)
ONE ENGINE TAXI DEPT	CONSIDER

AFTER START

ANTI ICE	AS RQRD
ECAM STATUS	CHECKED
PITCH TRIM	____ SET
RUDDER TRIM	ZERO

BEFORE TAKEOFF

FLIGHT CONTROLS	CHECKED (BOTH)
FLT INST	CHECKED (BOTH)
BRIEFING	CONFIRMED
FLAP SETTING	CONF____ (BOTH)
V1, VR, V2/FLX TEMP	_____ (BOTH)
ATC	SET
ECAM MEMO	TO NO BLUE
. AUTO BRK MAX	
. SIGNS ON	
. CABIN READY (◁)	
. SPLRS ARM	
. FLAPS TO	
. TO CONFIG NORM	
TAKEOFF RWY	____ CONFIRM (BOTH)
CABIN CREW	ADVISED
TCAS	TA OR TA/RA
ENG MODE SEL	AS RQRD
PACKS	AS RQRD

AFTER TAKEOFF / CLIMB

LDG GEAR	UP
FLAPS	RETRACTED
PACKS	ON
BARO REF	____ SET (BOTH)

APPROACH

BRIEFING	CONFIRMED
ECAM STATUS	CHECKED
LANDING DISTANCE	CONFIRM
SEAT BELTS	AS RQRD
BARO REF	____ SET (BOTH)
MINIMUM	____ SET (BOTH)
ENG MODE SEL	AS RQRD

LANDING

CABIN CREW	ADVISED
A/THR	SPEED/OFF
AUTOBRAKE	AS RQRD
ECAM MEMO	LDG NO BLUE
. LDG GEAR DN	
. SIGNS ON	
. CABIN READY (◁)	
. SPLRS ARM	
. FLAPS SET	

AFTER LANDING

FLAPS	RETRACTED
SPOILERS	DISARMED
APU	AS RQRD
RADAR	OFF/STBY
PREDICTIVE WINDSHEAR SYSTEM	OFF
ONE ENGINE TAXI ARR	CONSIDER
PACK 2	AS RQRD

PARKING

Y ELEC PUMP	OFF
APU BLEED	AS RQRD
ENGINES	OFF
SEAT BELTS	OFF
EXT LT	AS RQRD
FUEL PUMPS	OFF
PARK BRK and CHOCKS	AS RQRD
COMPANY MOBILES	ON
Consider HEAVY RAIN	

SECURING THE AIRCRAFT

BRAKE TEMPERATURE	CHECKED
ADIRS	OFF
OXYGEN	OFF
APU BLEED	OFF
EMER EXIT LT	OFF
SIGNS / NO SMOKING	OFF
APU AND BAT	OFF
Consider COLD WEATHER	

Source: Midkiff, Hansman, Reynolds (2009), picture source: slideshare.net

Example Flight

Taxi to RWY 26L via taxiways W2, D4, S8, S, B15

DO NOT TAXI ON RWY WITHOUT PRIOR PERMISSION BY ATC

MUC Apron 2 and Taxi Areas

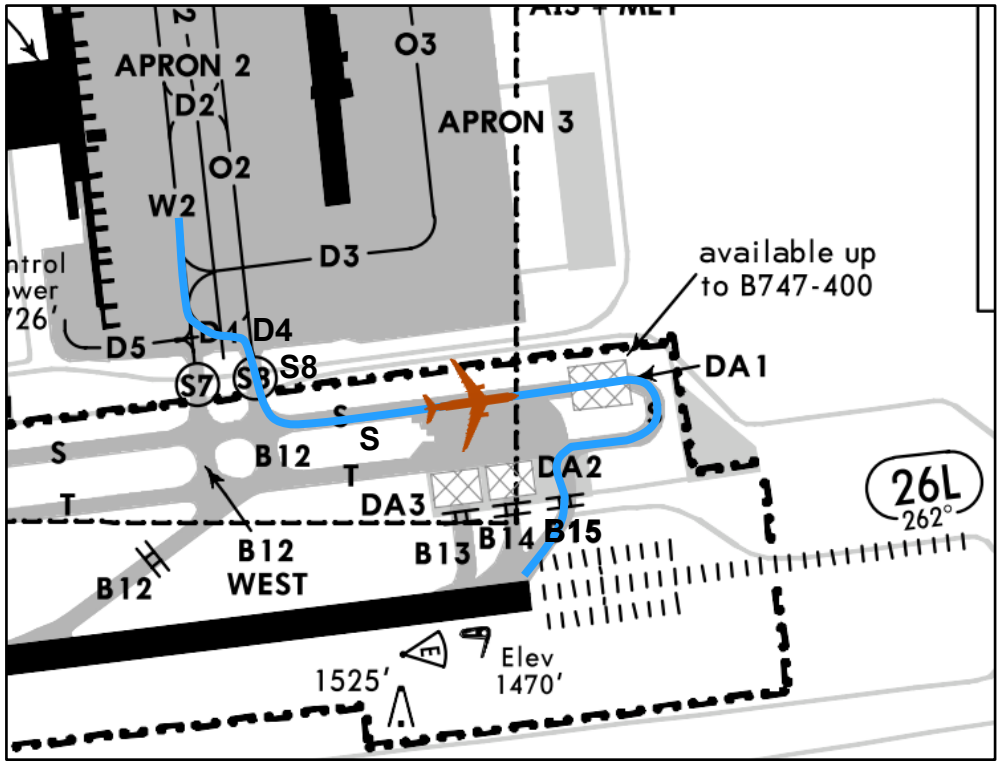




Image Source: Jeppesen,

7. Takeoff

Provided the flight deck and cabin crew have both **completely prepared the aircraft for takeoff**, the pilots may contact ATC ("Tower") to **receive permission to taxi on the runway and take off**.

- The aircraft is then either "***cleared for takeoff***" (rolling takeoff),
  - or must "***hold short of the runway***",
  - or is allowed to "***line up on runway and wait***".

- Once "*cleared for takeoff*" by ATC,
  - the pilots' roles of captain / first officer change to **Pilot Flying / Pilot Monitoring,**
  - the pilot flying pushes the engine throttle levers forward to **accelerate the aircraft**,
  - the crew monitors the **aircraft centerline tracking** and **engine parameters**,
  - the pilot monitoring calls out the **V-Speeds** ("V1", "Rotate," "V2").

- Once the aircraft is airborne,
  - it is **cleaned up** (gear up, slats / flaps up) after it reaches certain altitudes & speeds,
  - the crew observes **the traffic situation outside**.

Source: Midkiff, Hansman, Reynolds (2009)

Example Flight

Takeoff from RWY 26L

MUC RWY 26L

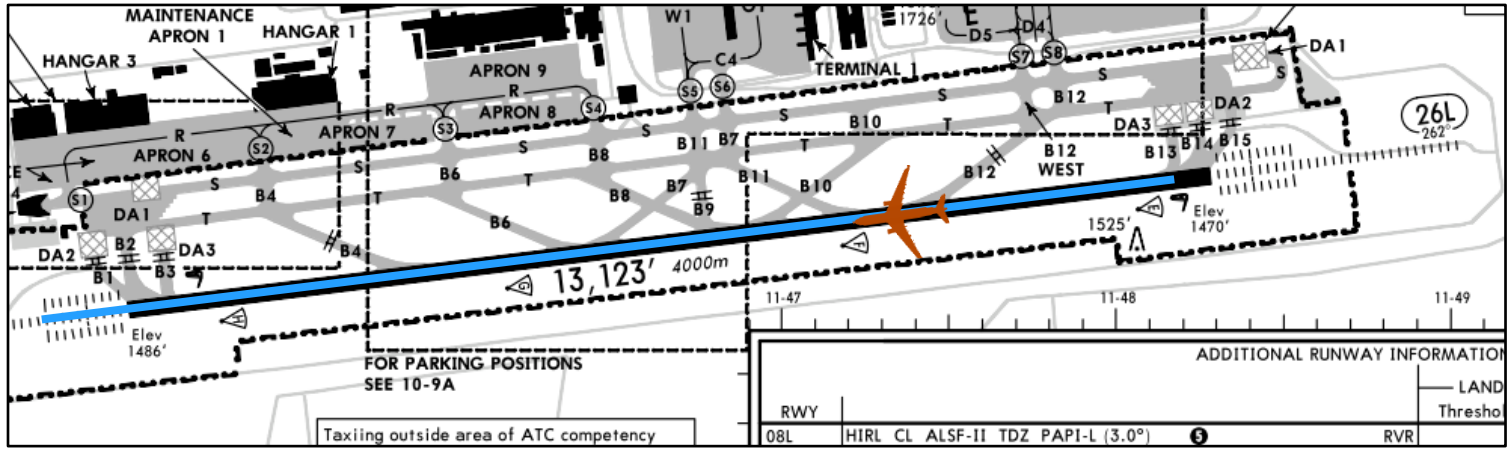




Image Source: Jeppesen

8. Terminal Area Departure

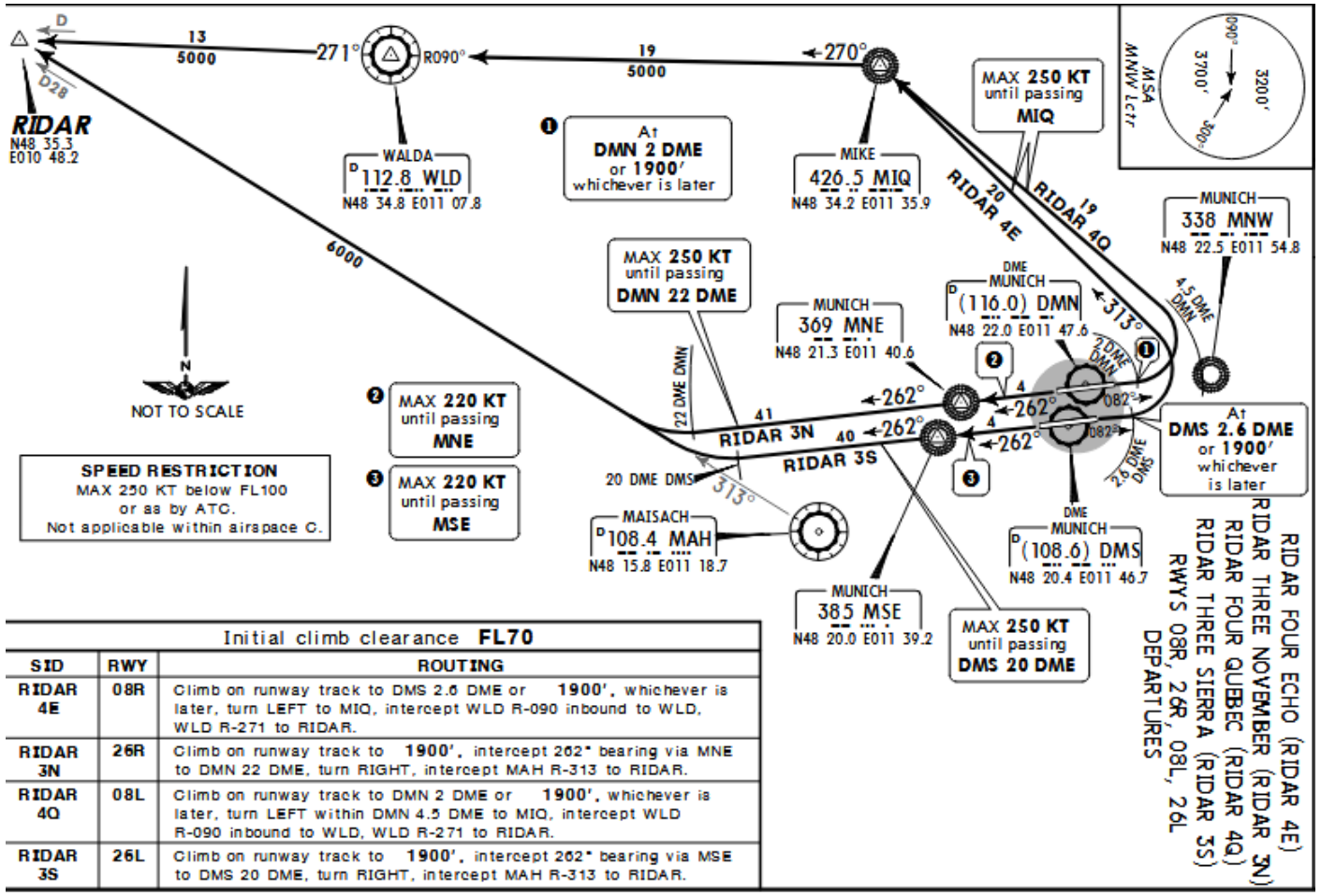
When clear of the immediate airport traffic area, the aircraft **is accelerated to maximum low-altitude climb speed** (usually 250 knots below FL100 or as advised by ATC).

- The pilot flying engages the **autoflight mode** of the aircraft.
- The aircraft follows the pre-defined **Standard Instrument Departure** (SID)
  - ➢ strategic autoflight mode
- …or complies with the **radar vectoring commands** as communicated by ATC
  - ➢ tactical autoflight mode
- The flight attendants typically **commence in-flight service** duties during climb-out unless the captain has briefed them otherwise, or the weather and/or ride conditions are prohibitive.
- Above FL100, the aircraft is accelerated to **the optimal climb speed**.

Source: Midkiff, Hansman, Reynolds (2009)

Example Flight

Follow SID



Initial climb clearance **FL70**

SID	RWY	ROUTING
RIDAR 4E	08R	Climb on runway track to DMS 2.6 DME or 1900', whichever is later, turn LEFT to MIQ, intercept WLD R-090 inbound to WLD, WLD R-271 to RIDAR.
RIDAR 3N	26R	Climb on runway track to 1900', intercept 262° bearing via MNE to DMN 22 DME, turn RIGHT, intercept MAH R-313 to RIDAR.
RIDAR 4Q	08L	Climb on runway track to DMN 2 DME or 1900', whichever is later, turn LEFT within DMN 4.5 DME to MIQ, intercept WLD R-090 inbound to WLD, WLD R-271 to RIDAR.
RIDAR 3S	26L	Climb on runway track to 1900', intercept 262° bearing via MSE to DMS 20 DME, turn RIGHT, intercept MAH R-313 to RIDAR.

Image Source: Jeppesen



9. Climb + 10. Cruise

At some point during the climb and during cruise, the cockpit crew checks the Flight Management System to **compare the optimal and maximum cruise altitudes** with the planned data and **desired cruise Mach number**

- ➢ flying speed vs. trip time as stipulated by the **Cost Index**

- The pilots use this information to **coordinate cruise altitude and speed with ATC**.
- The pilots **monitor the weather and turbulence conditions** and may request different routing and / or altitude clearances with ATC.
- The pilots **usually request route shortcuts with ATC** to reduce trip length and time, as well as minimize fuel consumption.
- **Passenger-related activities** during the climb include beginning the meal and beverages service, delivering any marketing announcements, and activating the entertainment system.

Source: Midkiff, Hansman, Reynolds (2009)

Example Flight

Routing during Climb



Flightplan:

```
   6  INPUD2N
0006  INPUD
....  .... ....
---------------
   5  Y102
0011  UPALA
....  .... ....
---------------
  12  Z109
0023  BAMKI
....  .... ....
---------------
```

EDDM

Image Source: SkyVector

Example Flight

Routing during Cruise – North Atlantic Tracks

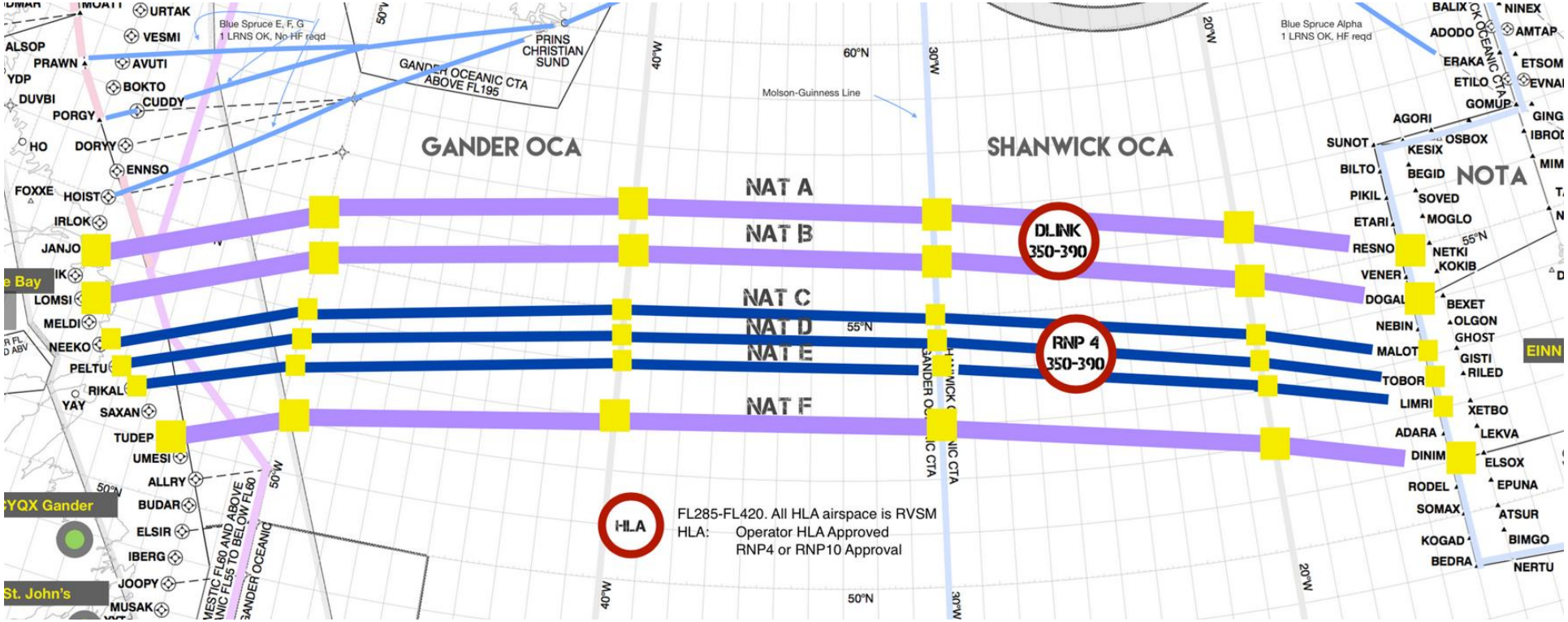


Image Source: SkyVector

11. Descent

Descent (leaving the cruise altitude) begins at the **“Top of Descent” (TOD)** point that is usually calculated by the FMS. It usually takes place with **about 20 to 30 minutes** remaining in the flight.

- The flight crew **requests to initiate the descent with ATC** (“*request descent*”).
- The **descent profile is determined by ATC restrictions**.
  - A continuous descent is rarely possible.
  - Future ATC systems envision Continuous Descent Approaches (CDA).
- The flight crew checks the **weather and operational situation at the destination airport** (via ATIS) and begins the approach and landing preparations accordingly.
  - Programming the navigation systems and FMS for the anticipated approach procedure and perform the approach briefing
- The cabin crew starts **preparing the cabin and galleys for landing**.

Source: Midkiff, Hansman, Reynolds (2009)

12. Terminal Arrival

Terminal area maneuvering generally begins when the aircraft descends below FL150 about 50 to 100 km from the destination airport.

- Usually, ATC assigns a **Standard Arrival (STAR)** to the aircraft that it is supposed to follow in order to reach the **Initial Approach Fix (IAF)**.

- ATC may also provide "**Radar Vectoring**," announcing heading, altitude, and speed commands to the pilots that steer the aircraft accordingly
  - ➢ tactical autoflight mode

- As the flight nears the position where it will commence final approach, ATC may issue precise information regarding **the actual runway used for landing** and **possible delays**.

- The flight crew **finalizes approach preparations** and informs the cabin crew and passengers about possible delays.

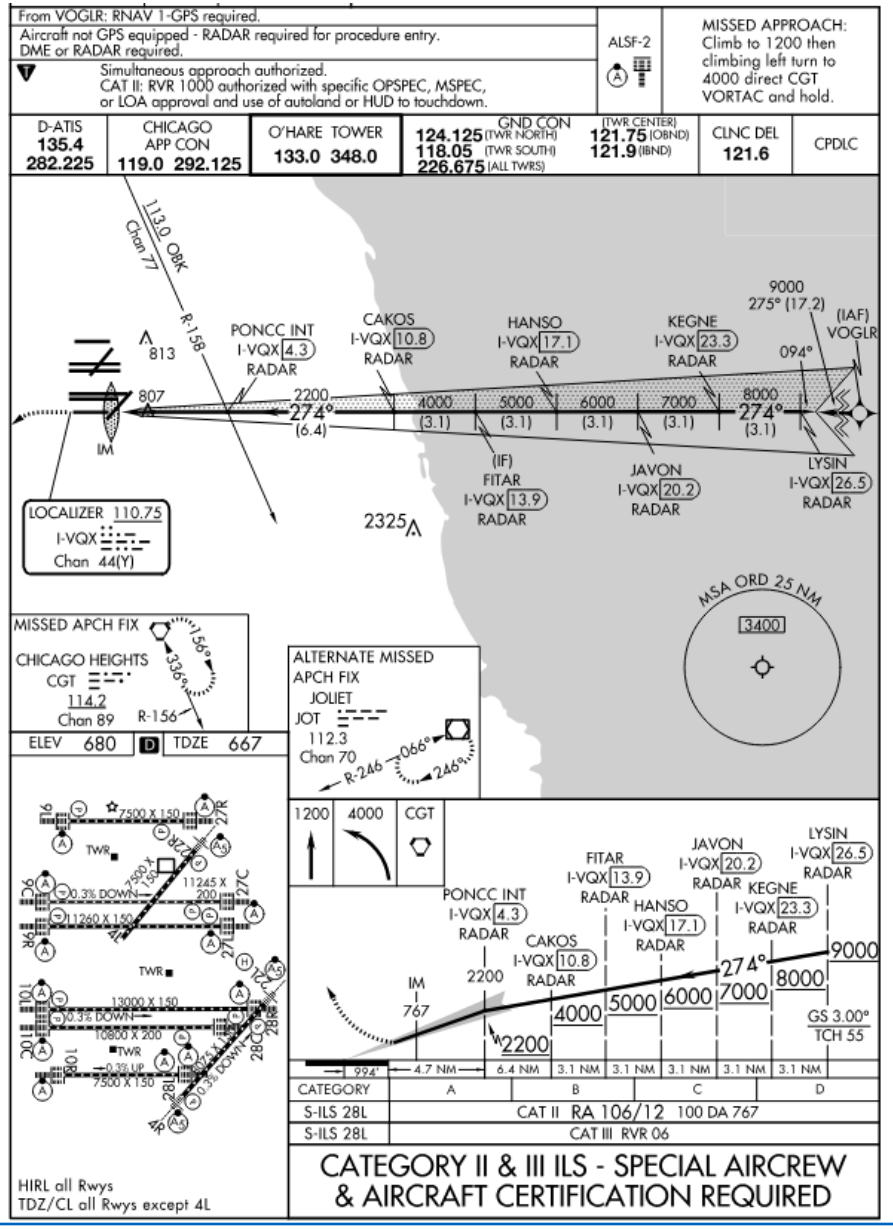
Source: Midkiff, Hansman, Reynolds (2009)

Example Flight

Follow



Image Source: Jeppesen



13. Final Approach

Once the aircraft arrives at the IAF, it commences **its final approach** to the **final approach fix (FAF)**, and eventually to the **runway** of the destination airport.

- Many runways at larger airports utilize **ILS (Instrument Landing System)** to provide guidance to aircraft during **IMC** (Instrument Meteorological Conditions).
- In **VMC** (Visual Meteorological Conditions), the pilots may request a **visual approach** with ATC.
- At some point during the vectoring of the aircraft provided by ATC, the aircraft will be "***cleared for the approach***," authorizing the cockpit crew to **start execution of the final approach procedure** as published in the aeronautical charts.
- During final approach, the pilots must **carefully examine the weather and especially the view and wind conditions**. Crosswind limitations of modern transport aircraft typically range from 25 to 35 knots.

Source: Midkiff, Hansman, Reynolds (2009)

Example Flight

Cleared for ILS approach RWY 28C



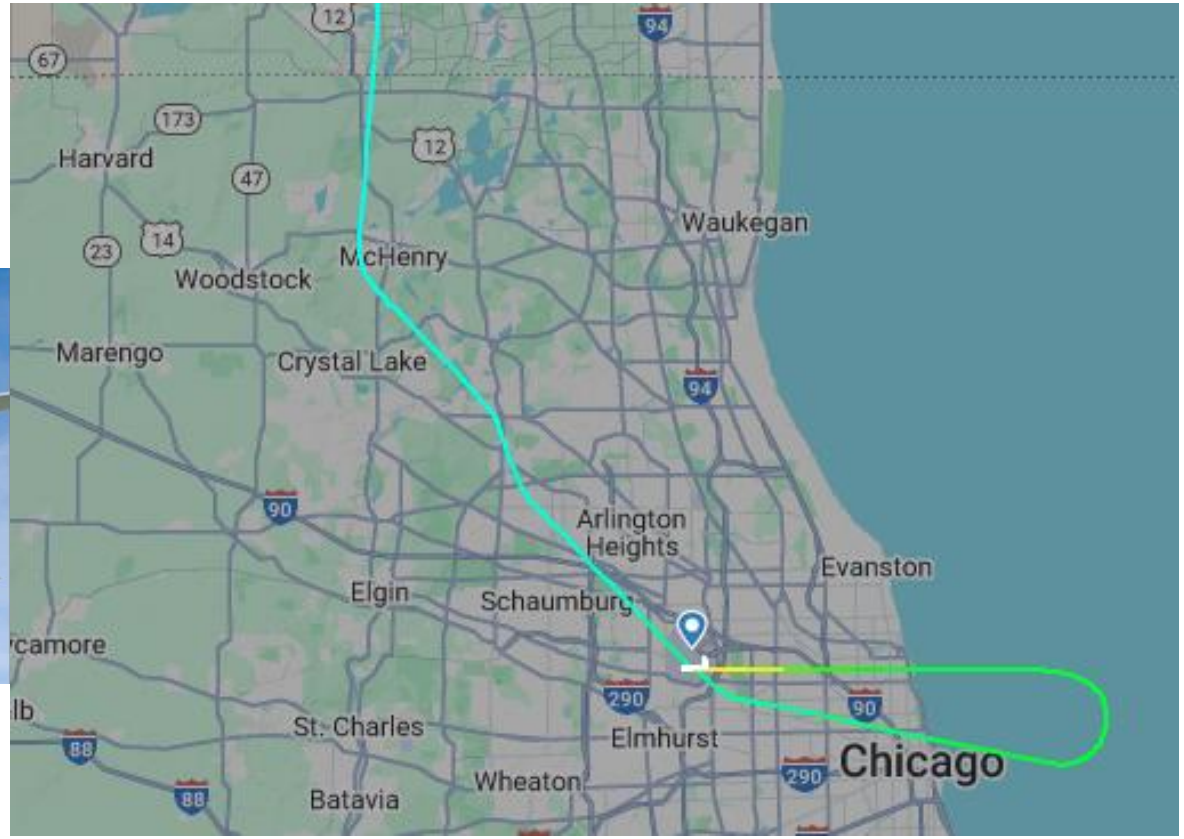


Image Source: Flightradar24

14. Landing and Rollout

After touching down on the runway, the pilot flying **uses a combination of reverse thrust, ground spoilers and wheel braking** to decelerate to taxi speed and vacate the runway.

- Reverse is always used, however full reverse may not be allowed at some airports or due to company policies.
- For safety and maximization of runway capacity, it is desirable to **minimize the runway occupancy times (ROT)** during rollout and the controller normally expects the crews to vacate at the **earliest safe opportunity**.
  - ➢ Use of "**high-speed taxiways**"
- As the aircraft slows down to taxi speed (10 – 20 kts), the captain and first officer **assume the taxi and communications tasks** as per normal ground operations.
- Once clear of the runway, the crew **reports any adverse wind or braking conditions** to ATC (Tower) and then **contacts ground control to request taxi-in instructions**.

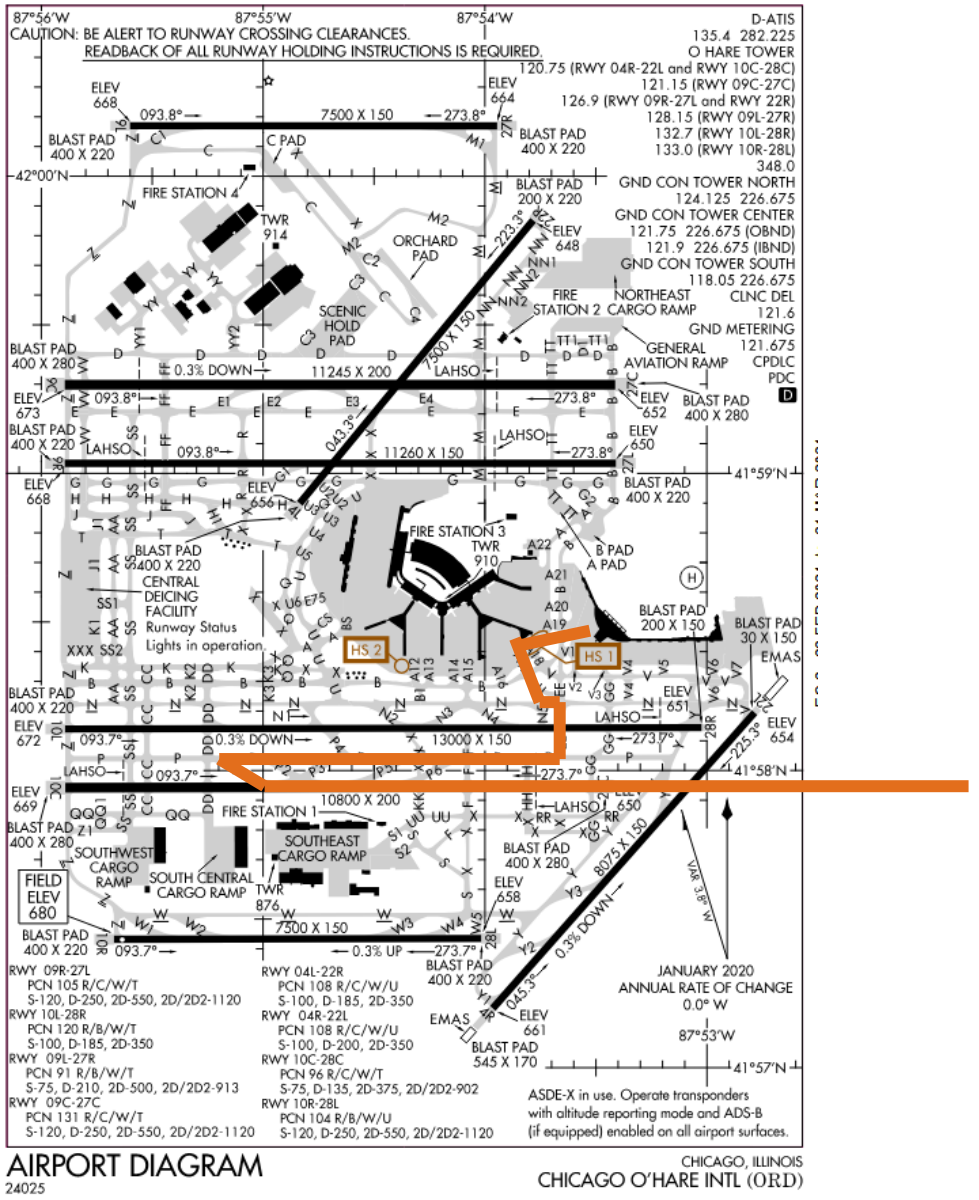
Source: Midkiff, Hansman, Reynolds (2009)

Example Flight

Landing on 28C



Image Source: Jeppesen



15. Taxi-In

The airport's **ground controller advises** the pilots which **taxiways** to take in order to arrive at the final parking stand or gate.

- The pilots use published **taxiway diagrams** of the destination airport to assist in the execution of the taxi clearances.
- Pilots must be **especially diligent** during ground operations at **airports with which they are unfamiliar or that are undergoing construction**.
- Before crossing an active runway, the pilots are **obliged to receive a clearance** to cross the runway by the **Tower**.
- Once clearance to the gate or final parking stand is received, **the captain taxis to the ramp area** and visually acquires the **marshaller** or **automated parking guidance system** provided at major airports.

Source: Midkiff, Hansman, Reynolds (2009)

16. Parking

Either a marshaller or an automated parking guidance system assist the pilots in **safely and precisely parking the aircraft**.

- In most cases, setting the parking brake and opening a cabin door trigger the **“IN” event**.
  - The “IN” time is **used to determine a number of metrics** including the duration of the flight (used to calculate flight crew compensation and legality for subsequent trips), on-time arrival reports, customs and immigration data, and company-specific performance monitoring and scheduling adjustments.

- Once a source of ground-based power and air supply is connected to the aircraft, the engines are shut down and the crew completes the “Engine Shutdown” checklist.

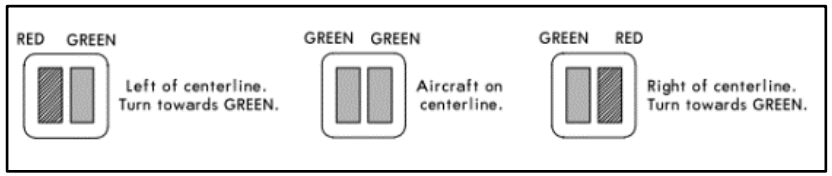
- The ramp agent **verifies the disarming of the doors** with the flight attendants and opens the designated exit door(s) to **start passenger disembarkation**.

- The flight crew **secures the cockpit and cabin** before departing the aircraft.

Source: Midkiff, Hansman, Reynolds (2009)

Example Flight

Parking the aircraft



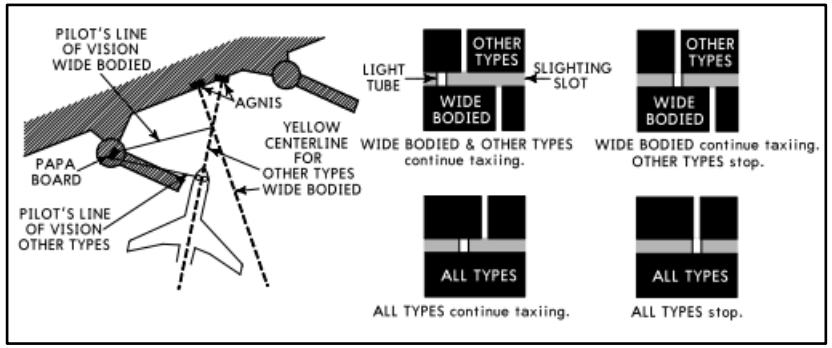




Image Source: Jeppesen

17. Post-Flight

Upon completion of post-flight duties at the aircraft, the cockpit crew accomplishes any required debriefing reports while the cabin crew make liquor and duty-free deposits, usually at the station operations.

- **TEC LOG, FLIGHT LOG** and **REM FUEL** completed, flight docs archived.
- **Debriefs / reports** are required by the cockpit crew in instances of…
  - a declared emergency of ATC violation, significant mechanical failures, fuel dumping, illness, injury or death of a passenger or crew member, passenger misconduct, overweight landing, possible diversion to an airport other than the originally intended destination, lightning strikes, high-speed aborts, near midair collisions, and a number of other situations **involving non-standard operations or issues**.

- Once all cockpit and cabin obligations are fulfilled, the **crew begins preparations for the next flight leg**.
  - In many cases, the crew must change the aircraft and relocate to the new departure gate where the planning and pre-flight duties are repeated.
  - If possible, the inbound and outbound crews should meet to **discuss any aircraft discrepancies or operational issues**.

Source: Midkiff, Hansman, Reynolds (2009)